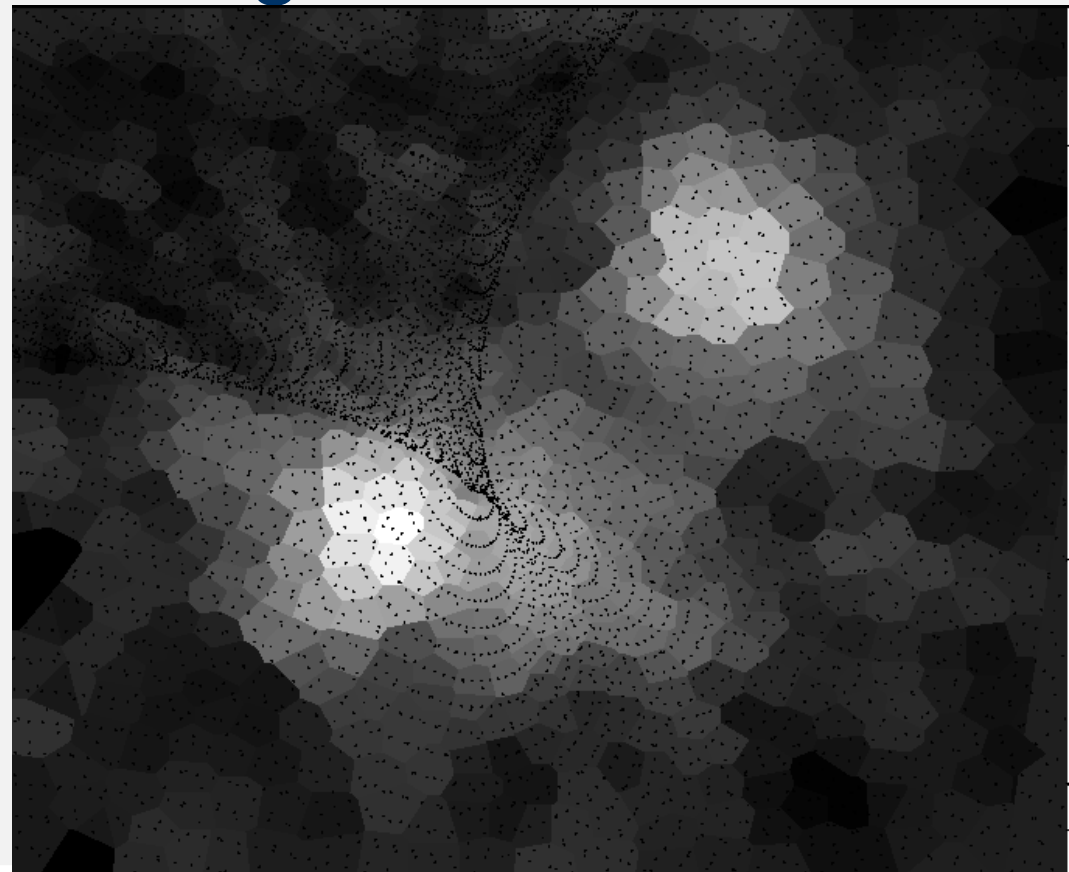




Adaptive Semi-linear Inversion: Sophisticated Lens Modeling of Large Strong-Lensing Data Sets

James Nightingale
Simon Dye





Introduction

- **What does sophisticated lens modeling measure?**
- **Adaptive Semi-linear Inversion**
 - Overview and formalism
 - Benefits
 - Data discretization – Unique and Unrelated Pixelizations
- **Strong Lens Analysis Pipeline**
 - Adaptive Semi-linear Inversion
 - Breakdown of multi-phase analysis
 - Example



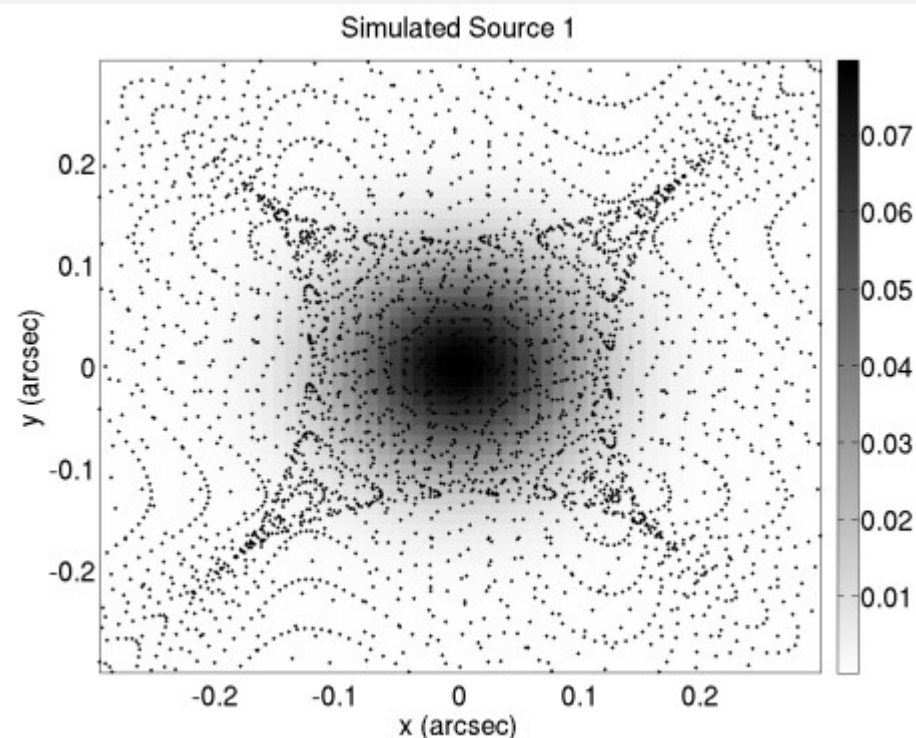
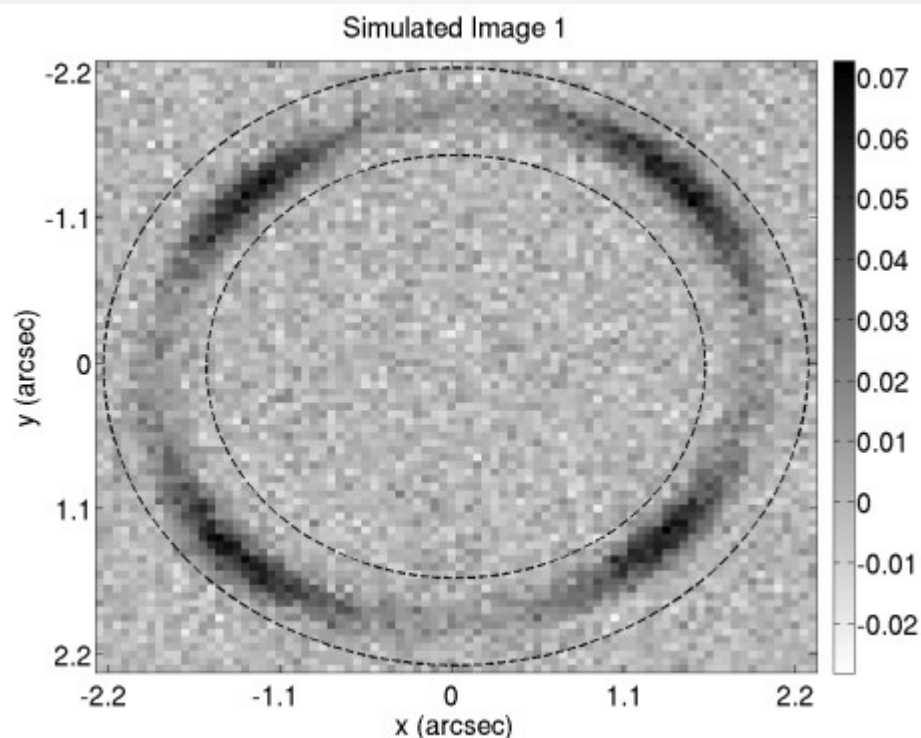
Adaptive SLI - Overview

- **Adaptive SLI has two primary goals:**
 - 1) Accurately perform modeling of complex, degenerate, lens models with just strong long data.
 - 2) Do this in a fast, efficient and streamlined manner allowing analysis to be performed on the largest data sets possible.



Square Semi-linear Inversion (SLI)

Simulated Image 1



Singular Power-Law Ellipsoid (SPLE) Lens model:

- (x,y) offsets = 0

- $q = b/a = 0.8$

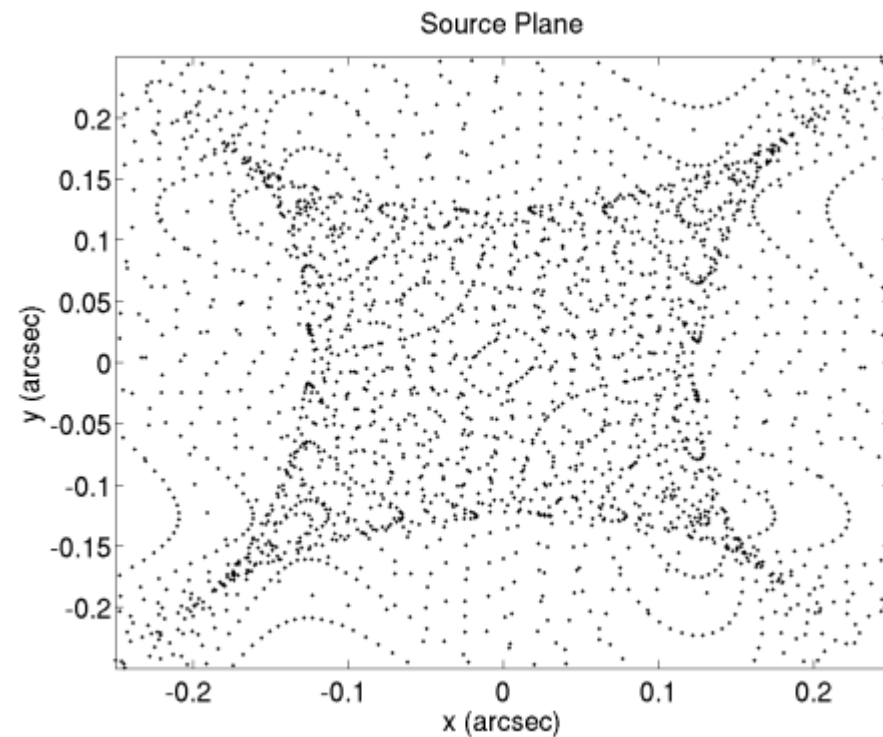
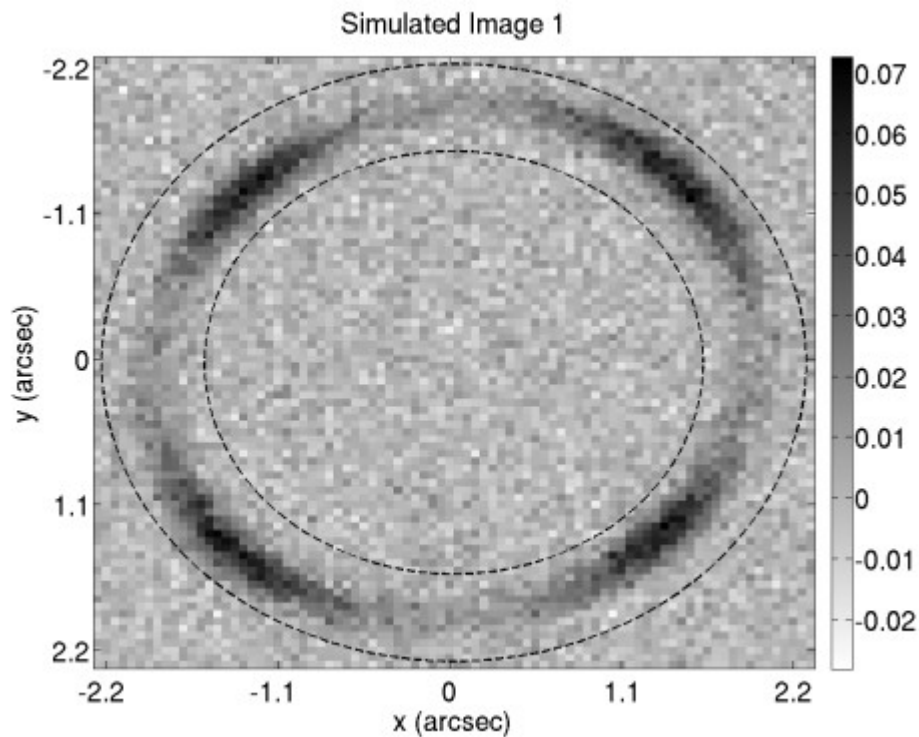
- $\sigma = 285 \text{ km/s}$

- $\Phi = 45^\circ$

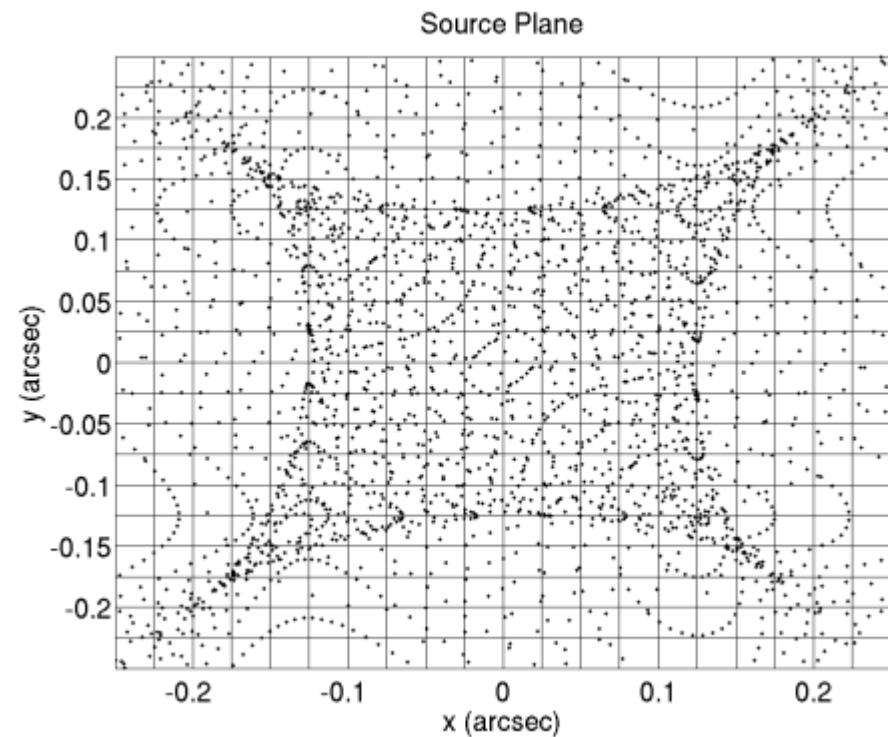
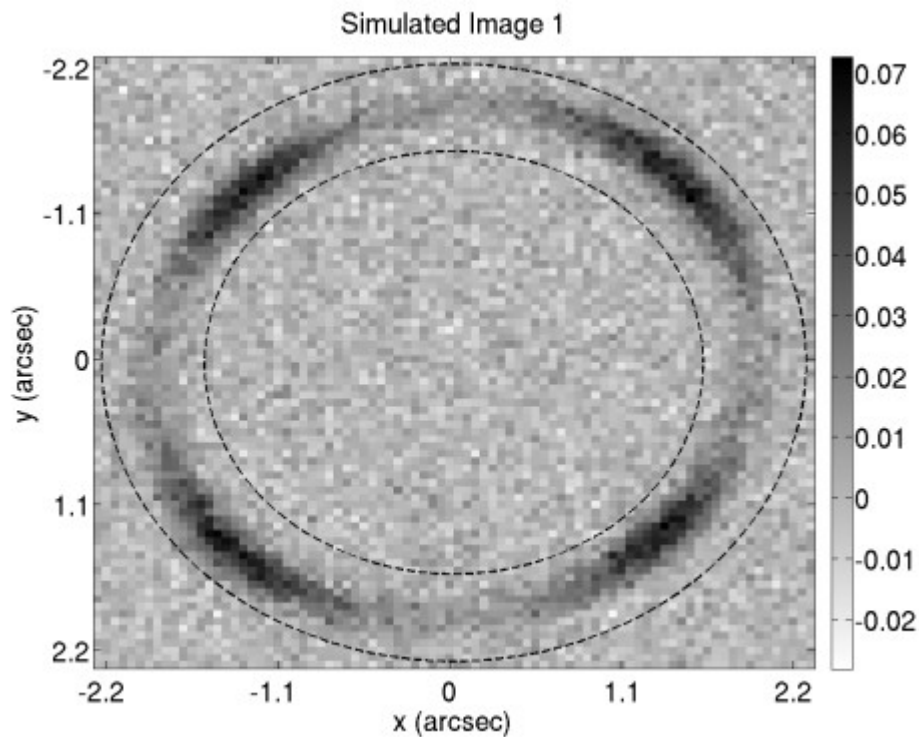
- $\alpha = 2.0 \text{ (SIE)}$

$$\rho(r) \propto r^{-\alpha}$$

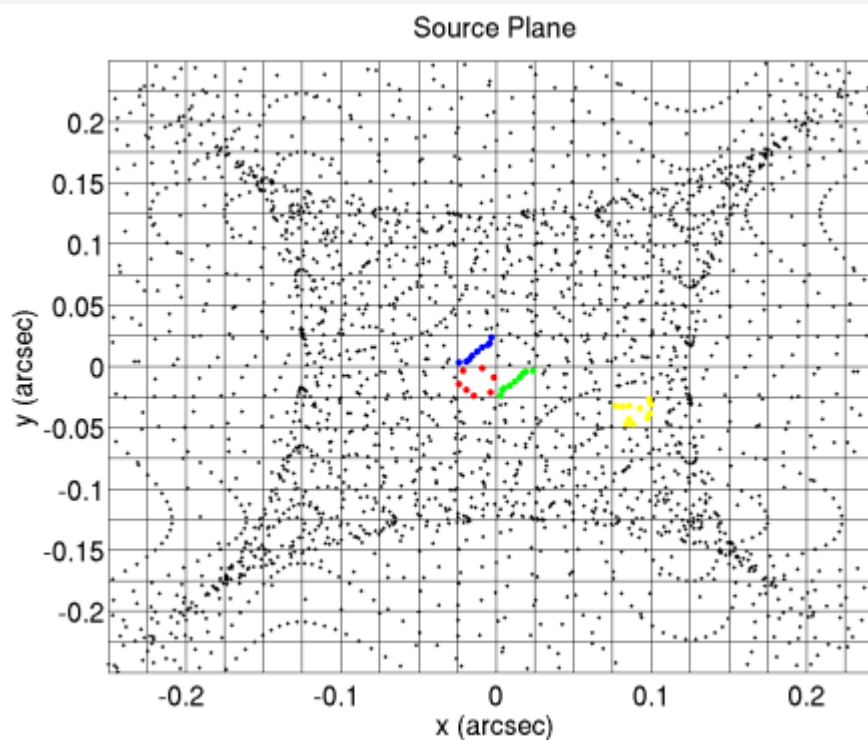
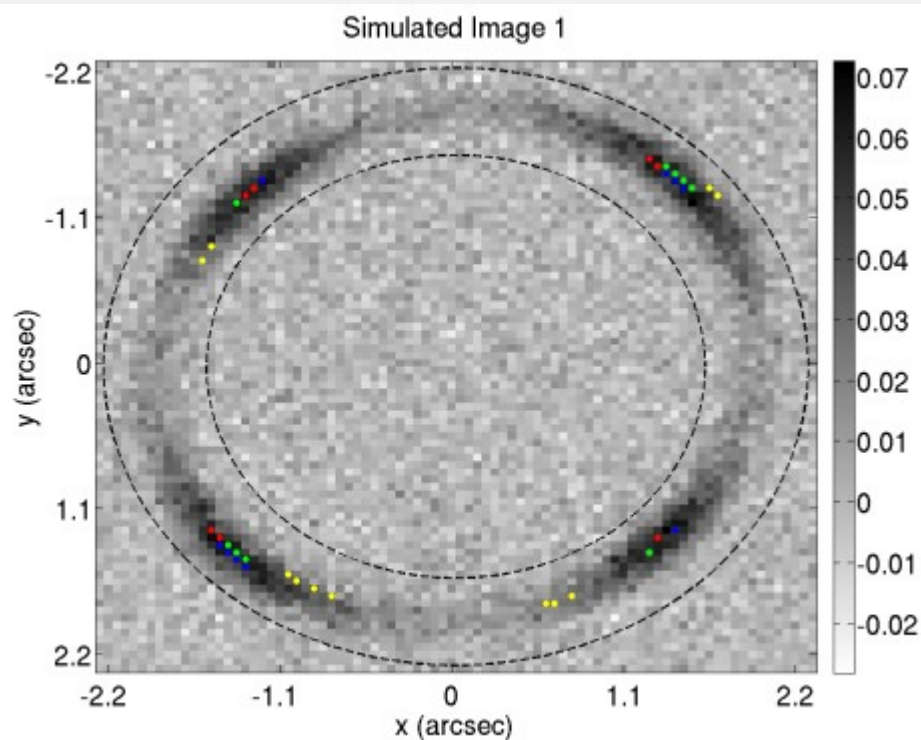
Step 1 – Trace Image Pixels to Source Plane



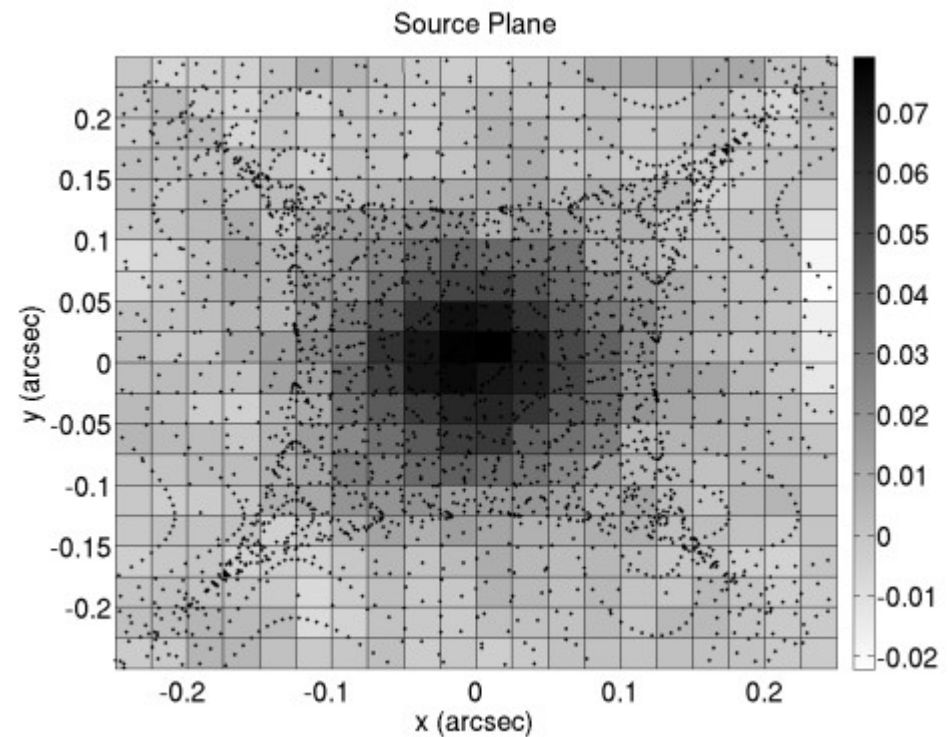
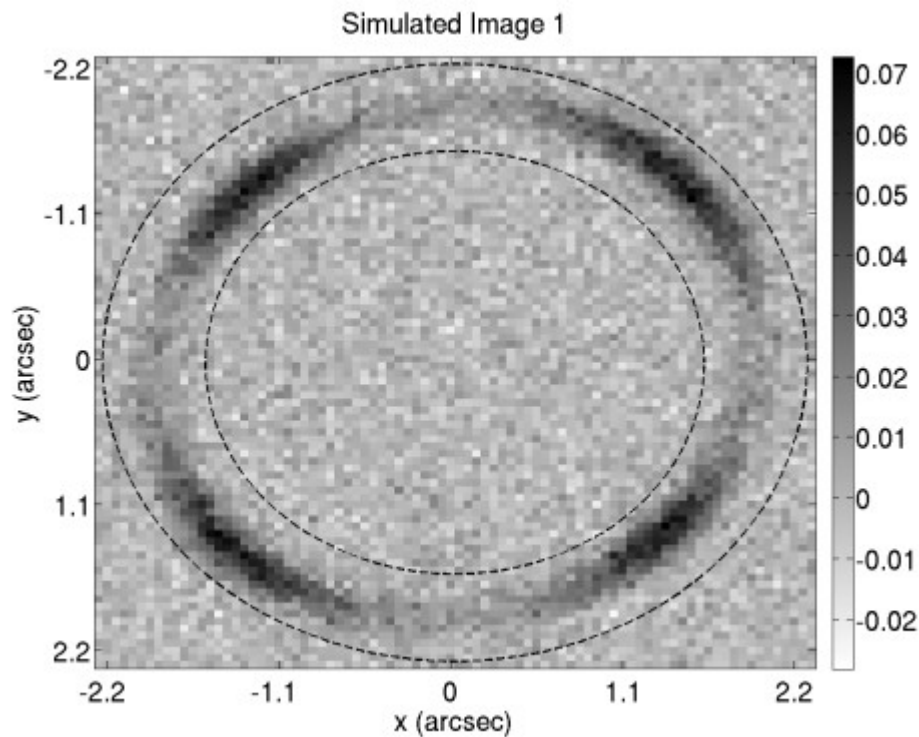
Step 2 – Overlay Square Grid



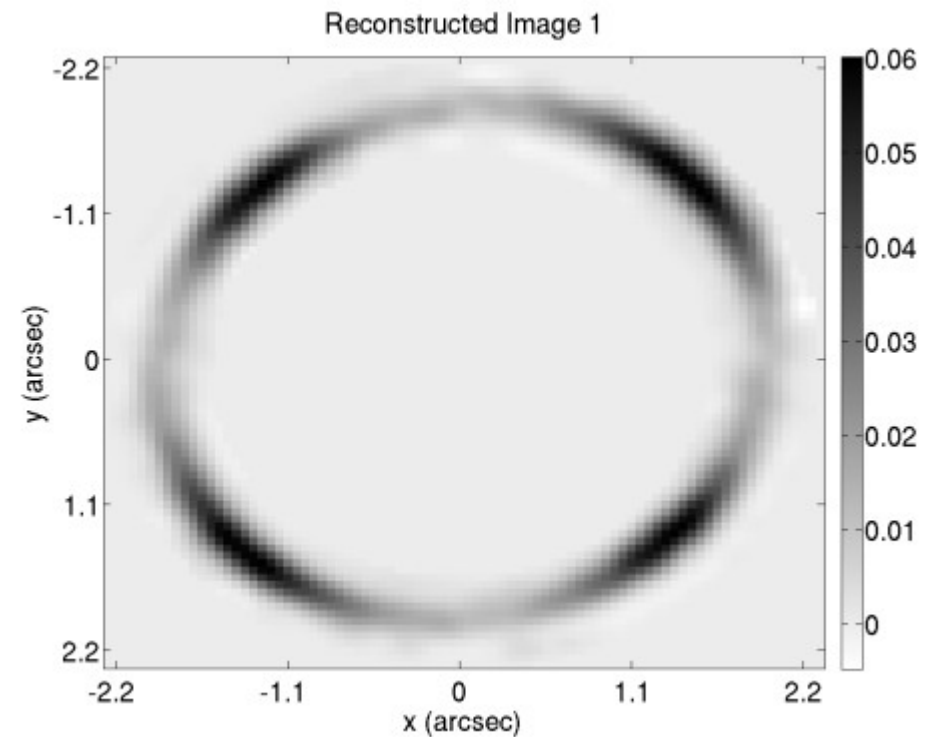
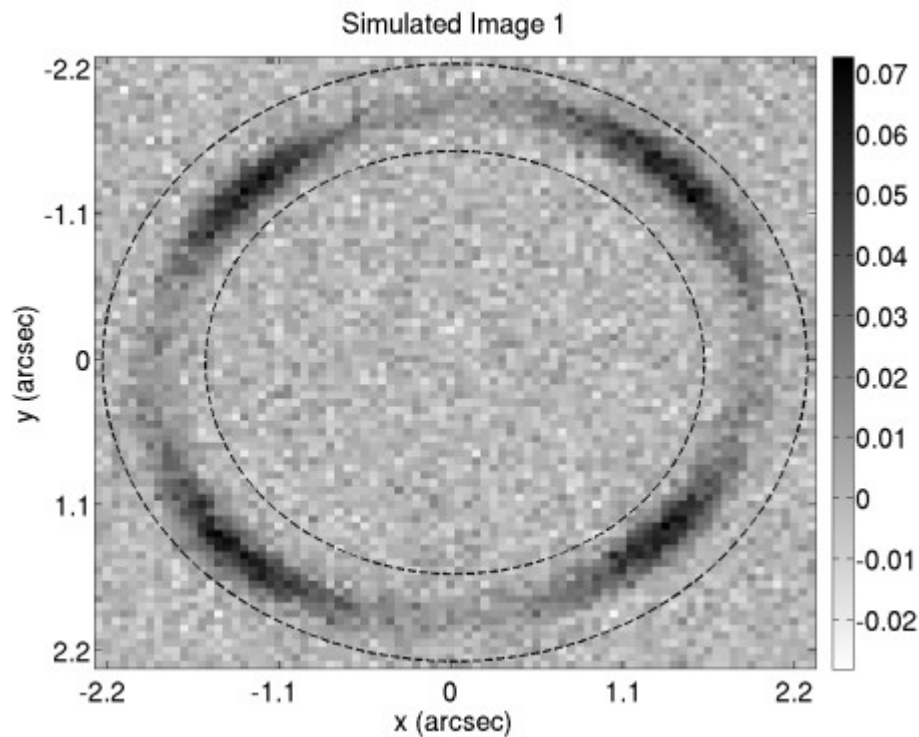
Step 3 – Image / Source Pixel Mappings



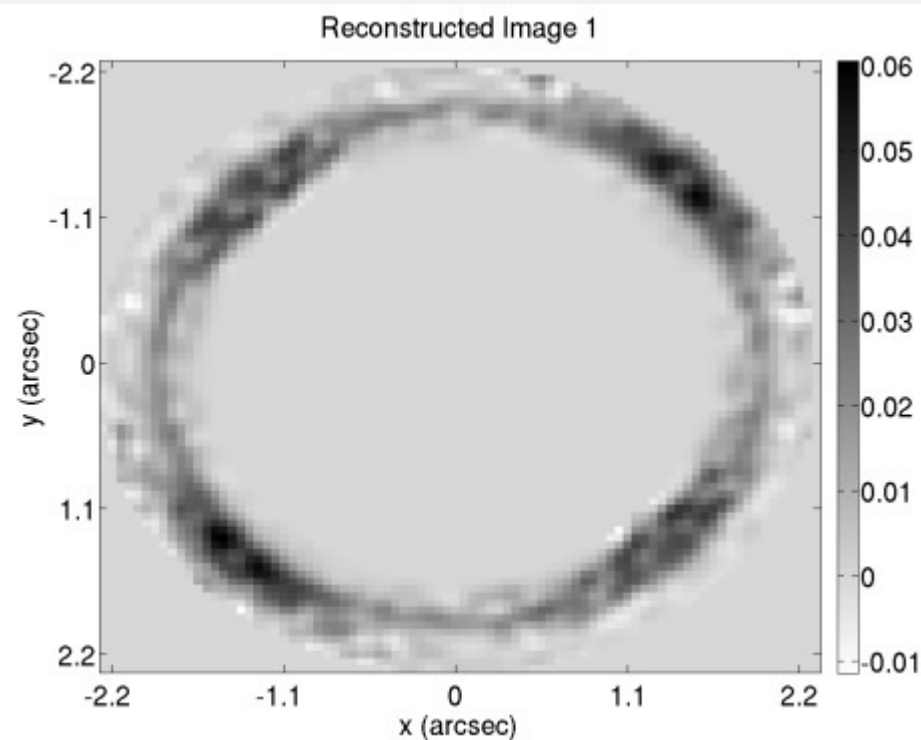
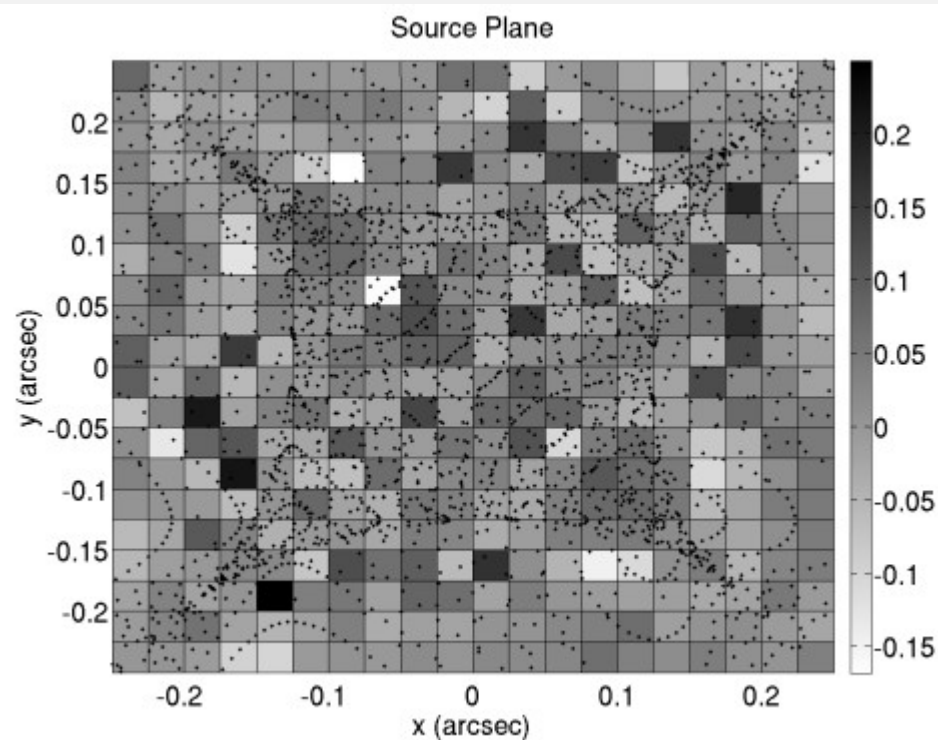
Step 3 – Reconstruct Source via Matrix Inversion



Step 4 – Compare Observed and Reconstructed Images



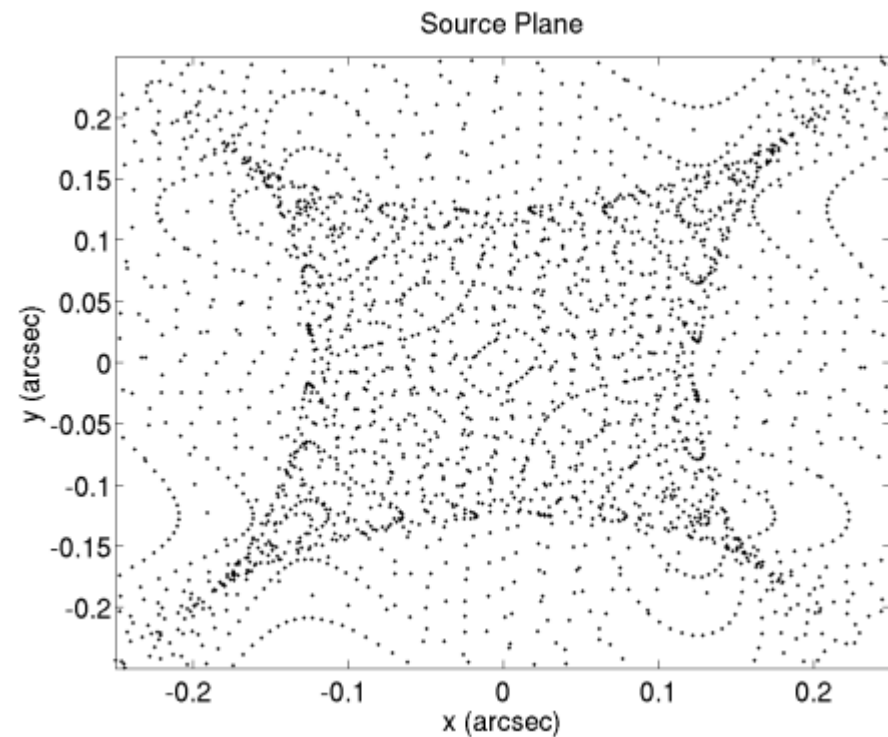
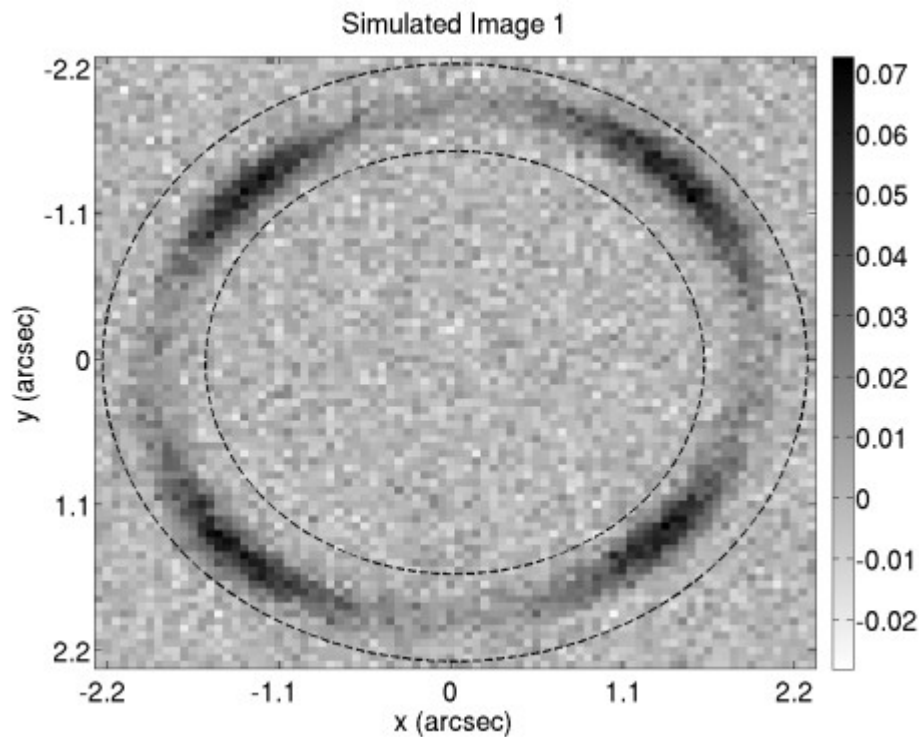
Example – Poor Lens Model



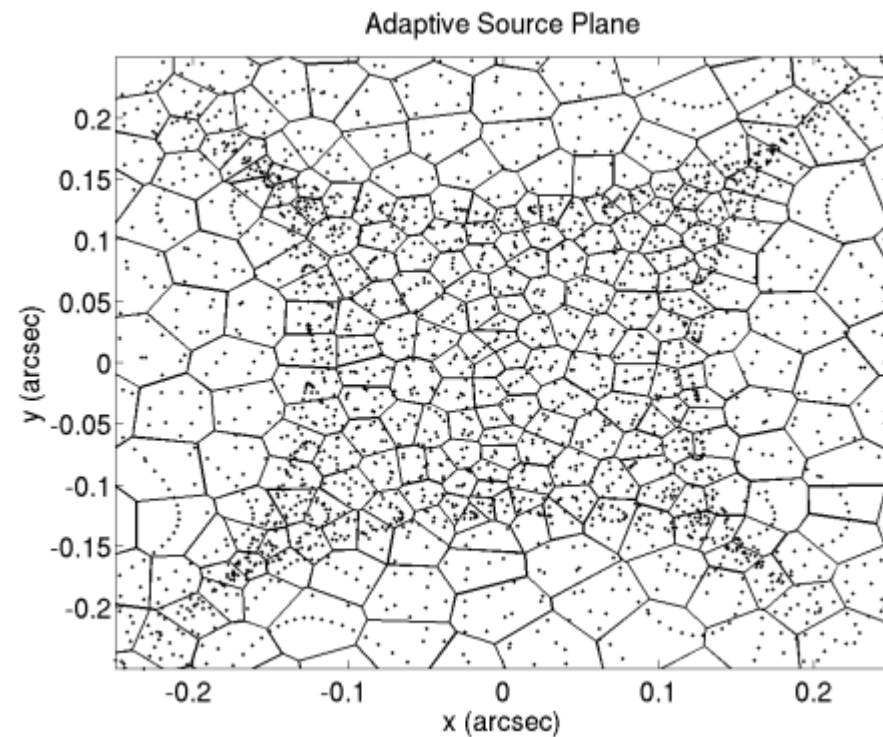
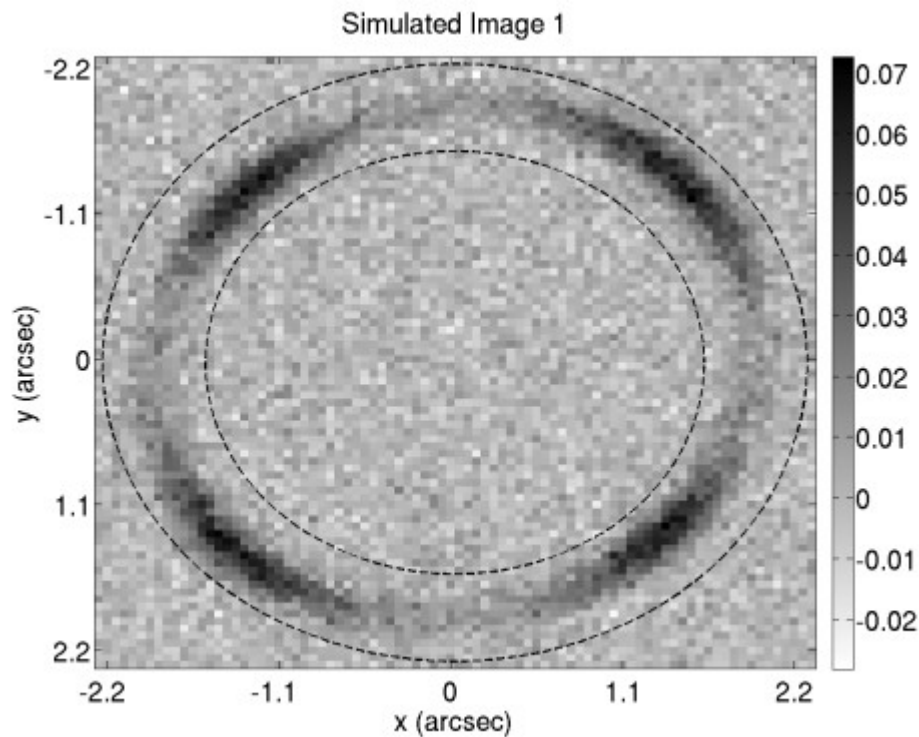


Adaptive Semi-linear Inversion (SLI)

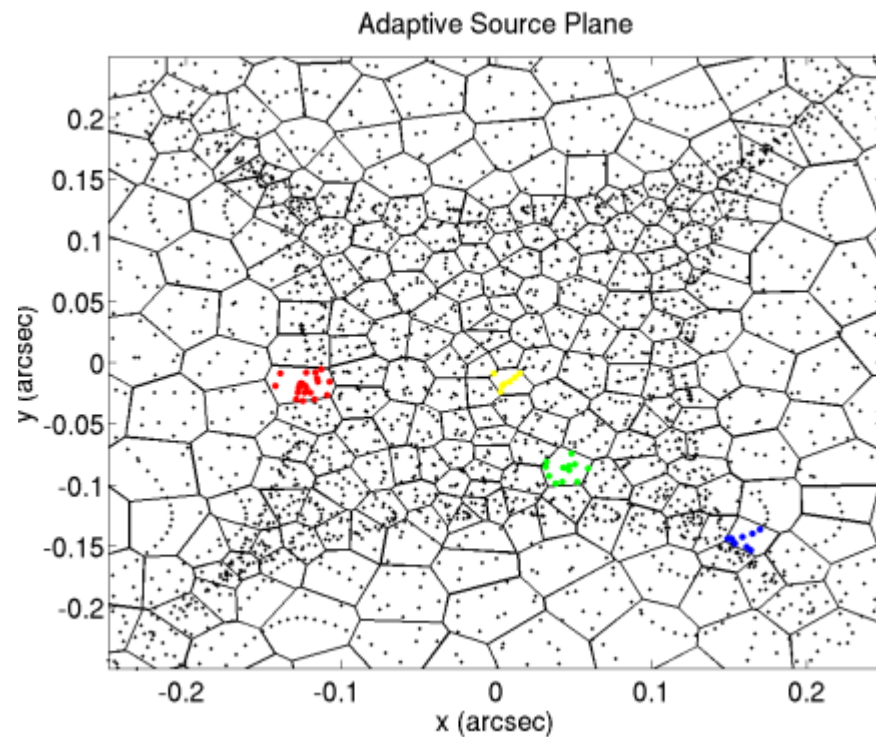
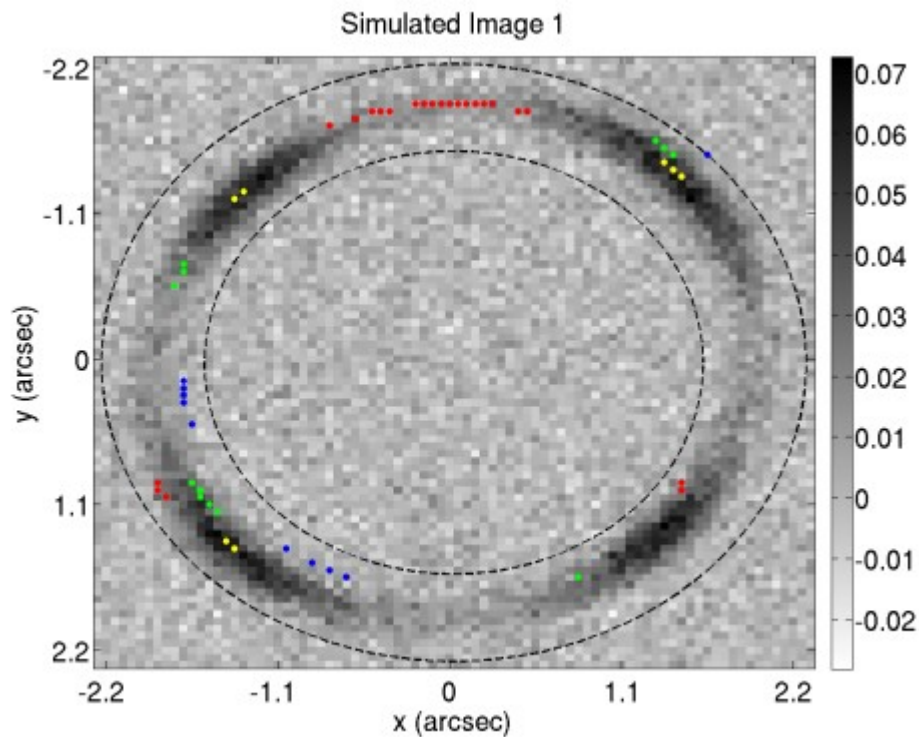
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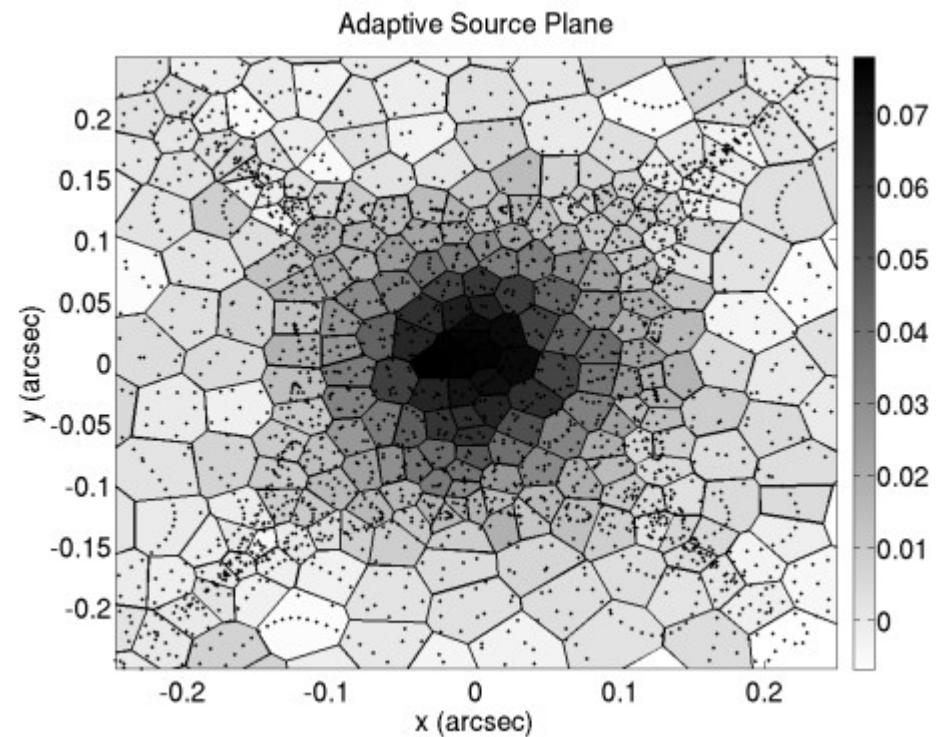
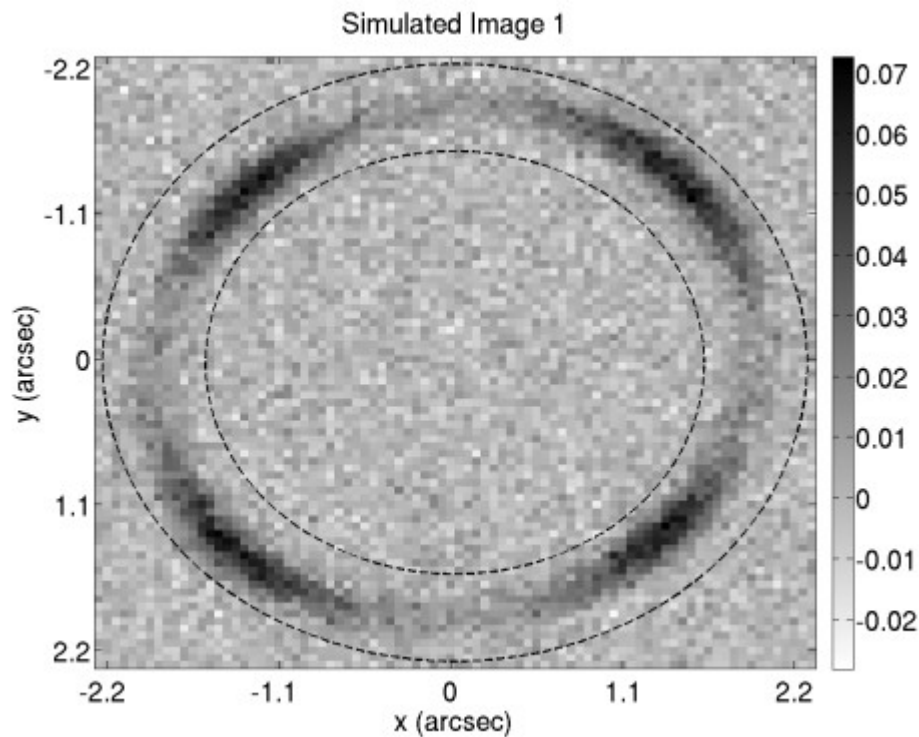
Step 2 – Input Into h-means Clustering Algorithm



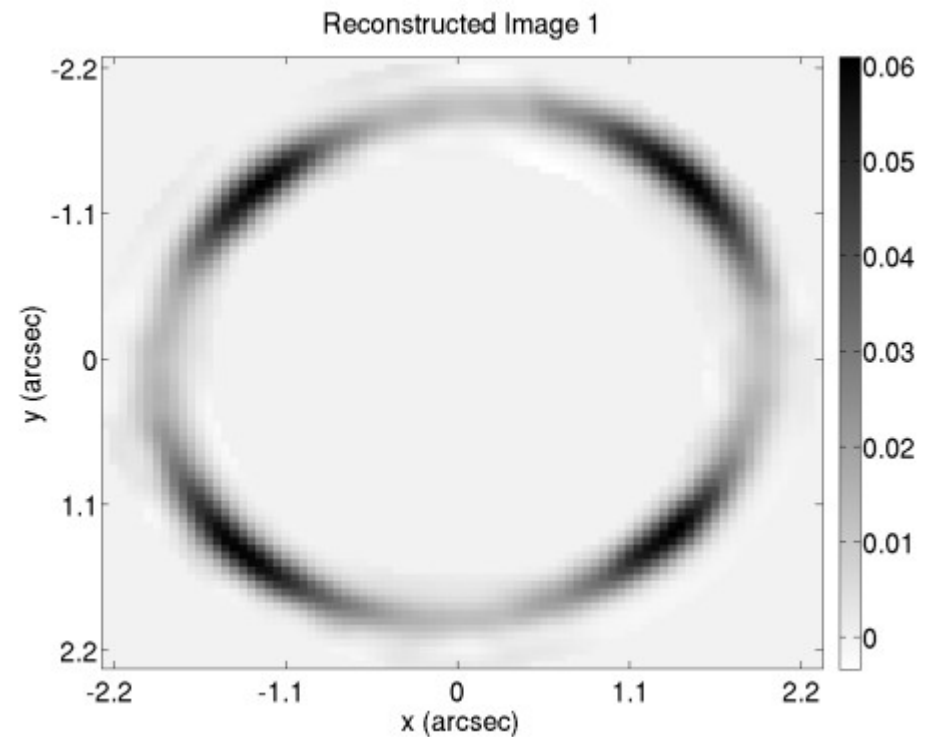
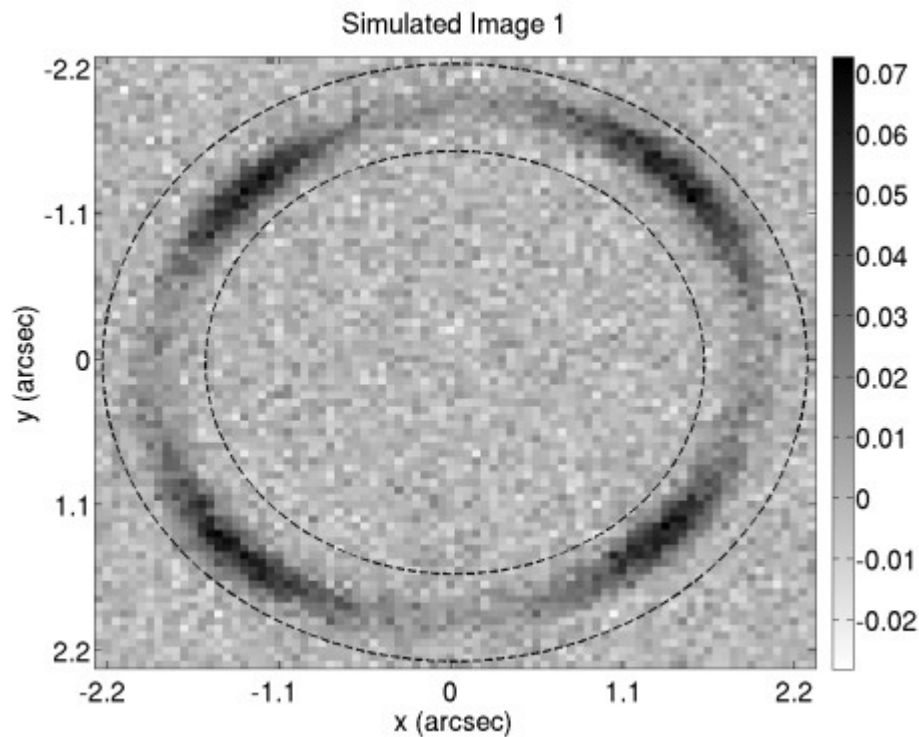
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- **Nearest-Neighbor regularization scheme used which penalizes non-constant solutions.**
 - Forces a physical source reconstruction.
 - Prevent over fitting to image noise.
- **Maximize Bayesian Evidence (Suyu et al. 2006) to set level of Regularization for every lens model.**



Determining the Lens Model

- **Every trial lens model returns a reconstructed image**
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- **Bayesian wrapper accounts for assumed regularization**
 - Converts χ^2 and regularization to a global evidence ε .
 - Maximize evidence ε -> most likely lens model.
- **Standard non-linear search problem.**
 - Find combination of lens parameters (x , y , σ , q , Φ , α) which maximize ε .
 - MultiNest (Feroz et al. 2009) algorithm.
 - Uses nested sampling technique to explore parameter space.



Benefits of Adaptive SLI

- **Pixelization derived from lens model magnification**
 - Lower source resolution required -> faster inversion.
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 - Easy to ensure all image pixels trace within source plane -> fix the NDOF (discussed later).

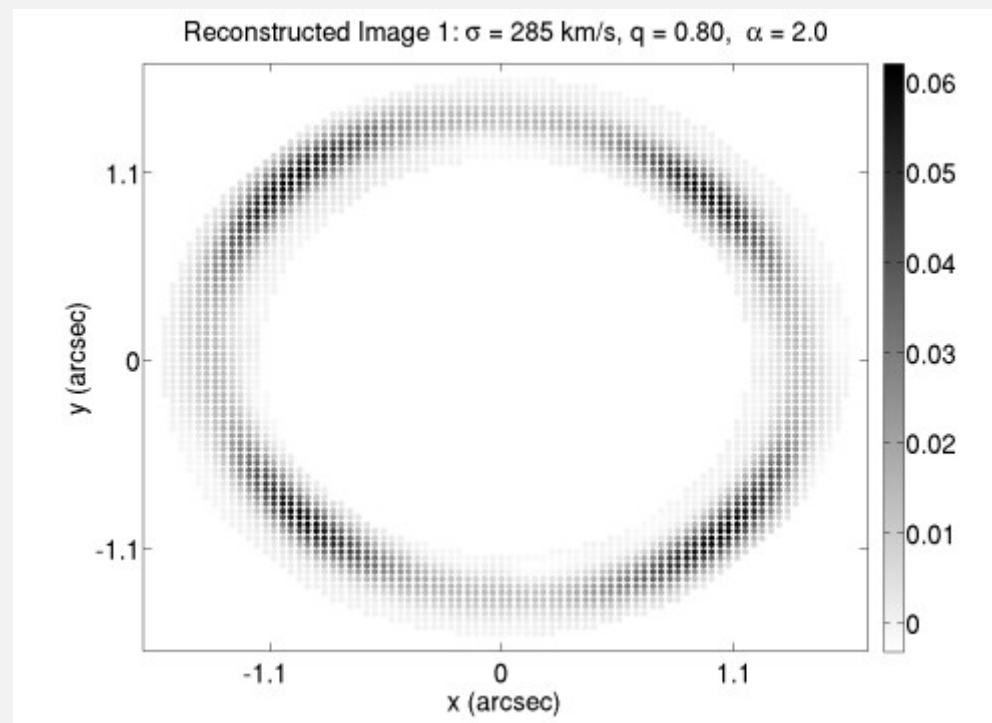


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 - Easy to ensure all image pixels trace within source plane -> fix the NDOF (discussed later).
- **Creates a *Unique* and *Unrelated* pixelization for every lens model**
 - Every pixelization is derived from a **random initialization**.
 - Data discretization is different for every inversion.

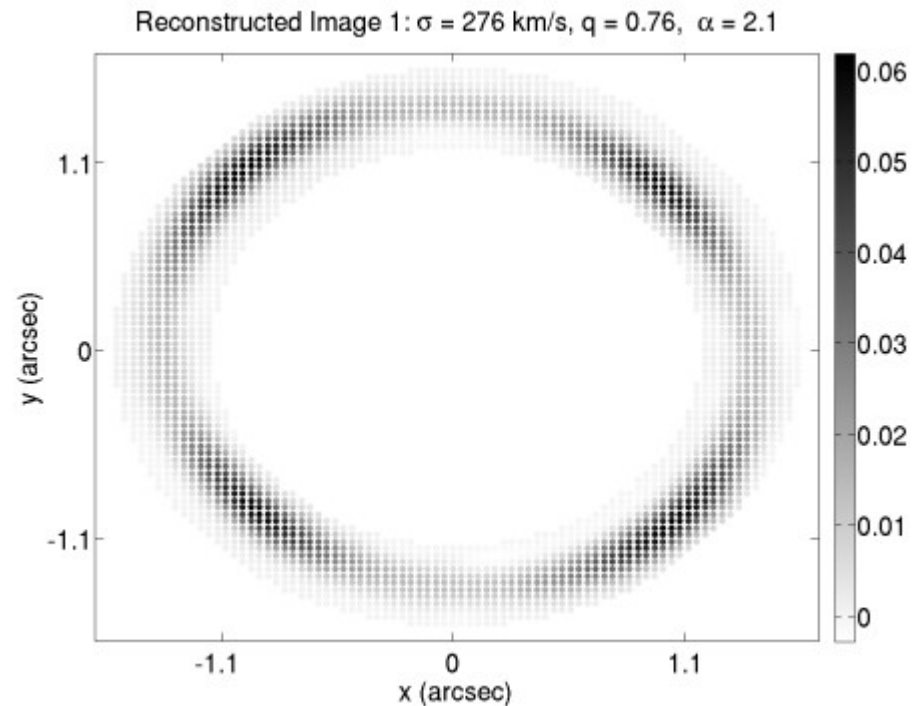
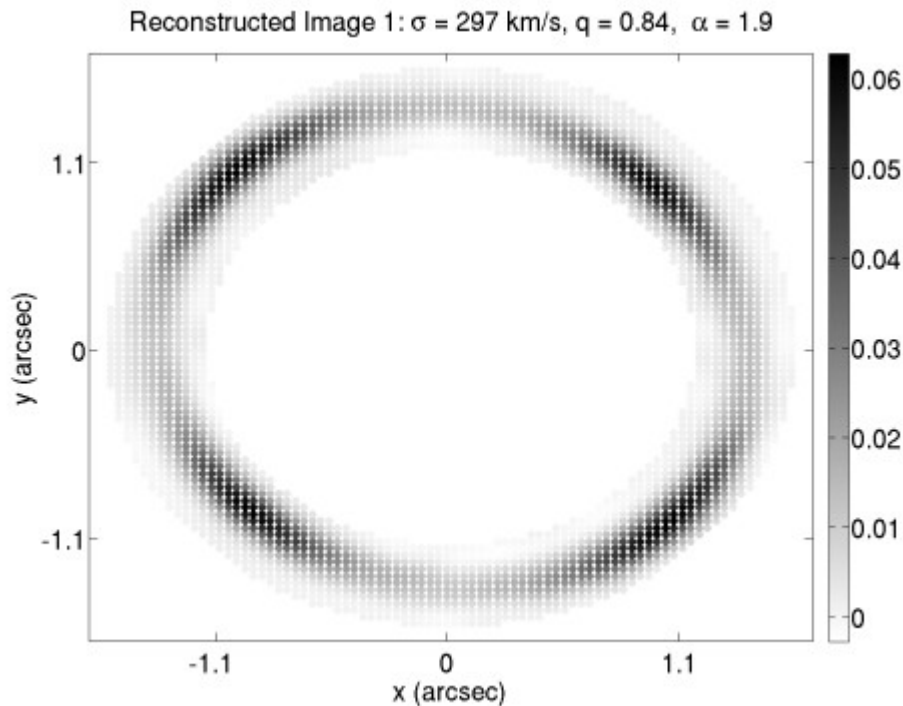
Fitting a Power-Law (PL) Lens Model

- Square SLI successfully fits a SIE lens model ($\alpha = 2$).
- **Fails to fit a PL lens model $\rightarrow \alpha$ free parameter.**
 - Degeneracy between (mass) σ and density slope α . (and q)



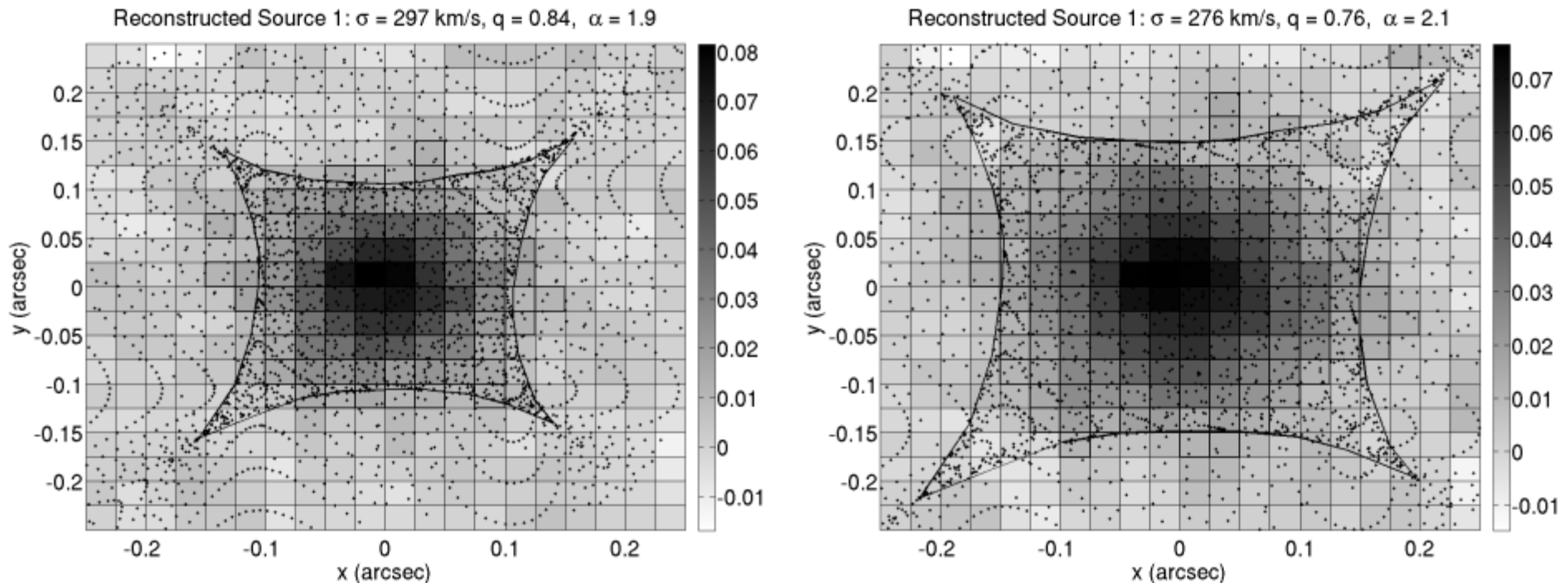
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Bias 1 – Varying NDOF

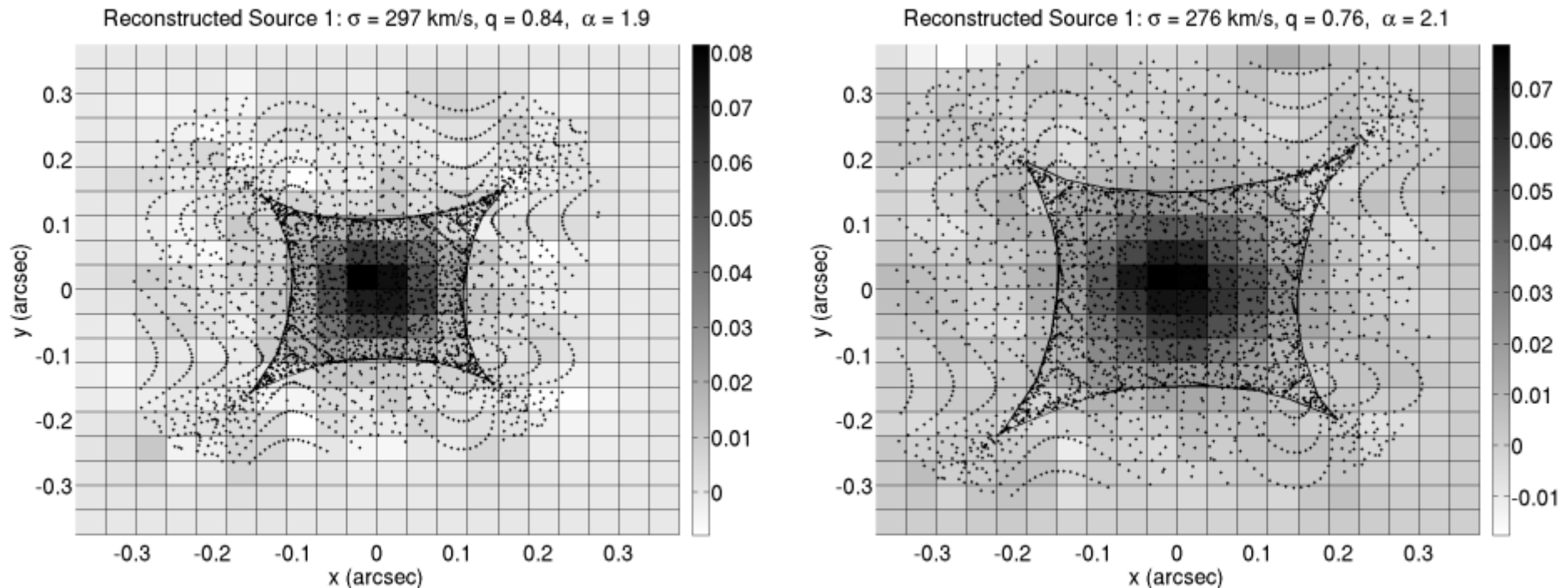
- **Degeneracy – Gives isotropic source scaling**



- **Source expands -> Fewer image pixels trace into source plane.**
- **Solutions biased to those which minimize the number of traced image pixels -> Must fix NDOF.**

Fix NDOF

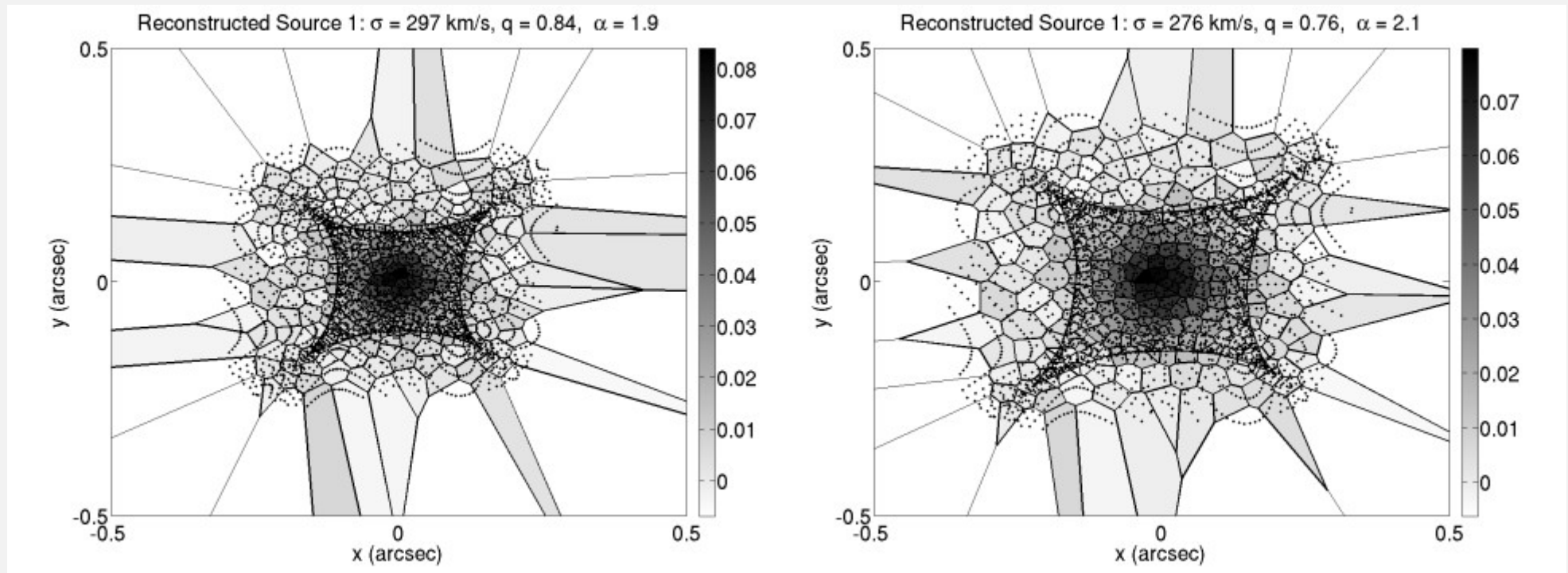
- **NDOF must be fixed for every possible lens model.**



- Large source plane, narrow priors and tightened image mask.
- inefficient resolution, pixels constrained by just regularization and varying effective resolution.

Fix NDOF

- **NDOF must be fixed for every possible lens model.**



- Can set Adaptive SLI source plane to arbitrarily large size.
- Circumvents all previous problems



Data Discreteness Bias

- **Even with the NDOF fixed, Square SLI could not accurately fit a PL lens model.**
 - Different lens model calculated if inversion setup is changed.
 - Results varied with source plane resolution, size and position.



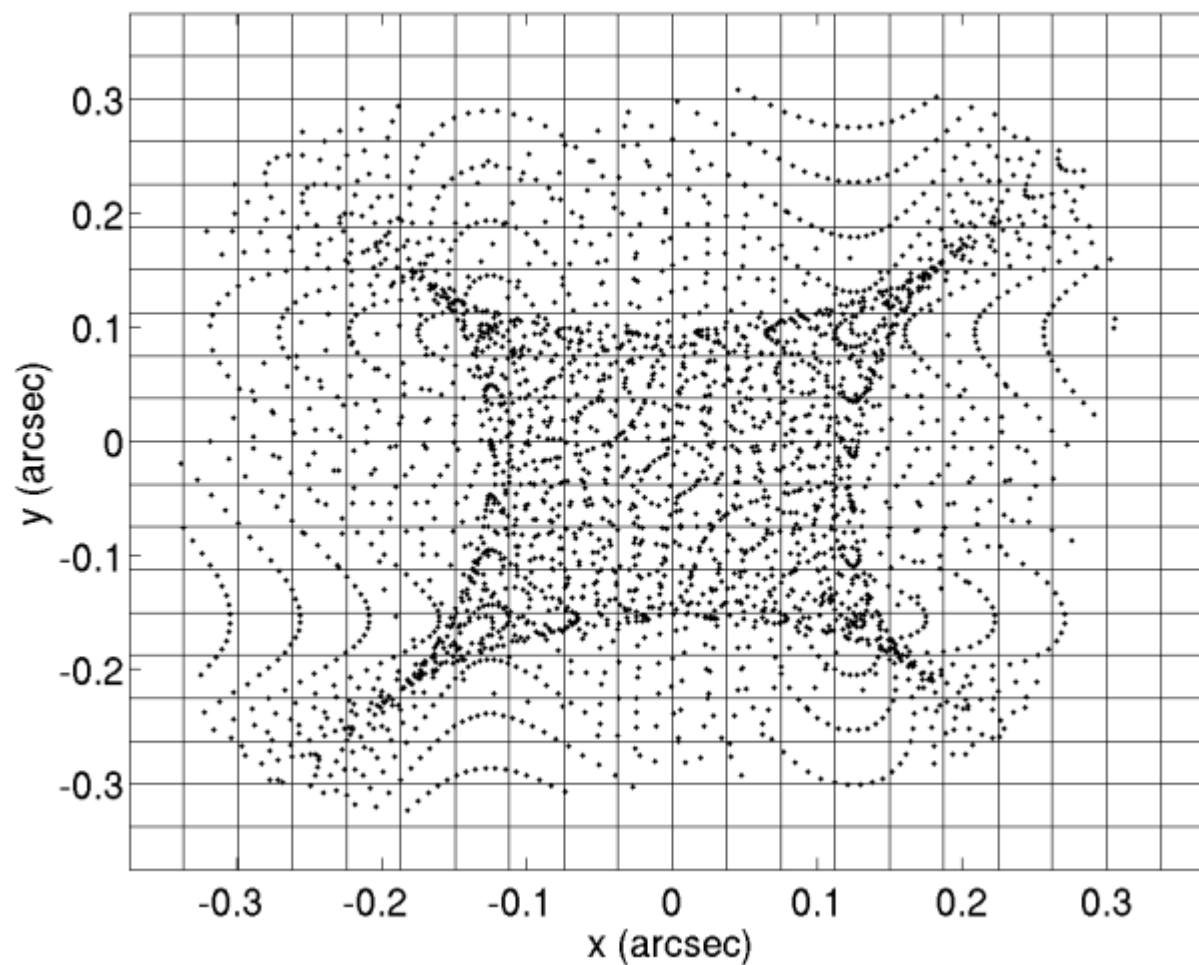
Data Discreteness Bias

- **Even with the NDOF fixed, Square SLI could not accurately fit a PL lens model.**
 - Different lens model calculated if inversion setup is changed.
 - Results varied with source plane resolution, size and position.
- **Investigation -> why do results change with setup?**
 - Fit 9 PL lens models to simulated image shown previously.
 - All 9 inversions identical, except source plane position.
 - Phase shift each grid by a fractional pixel interval.



Phase Shifts

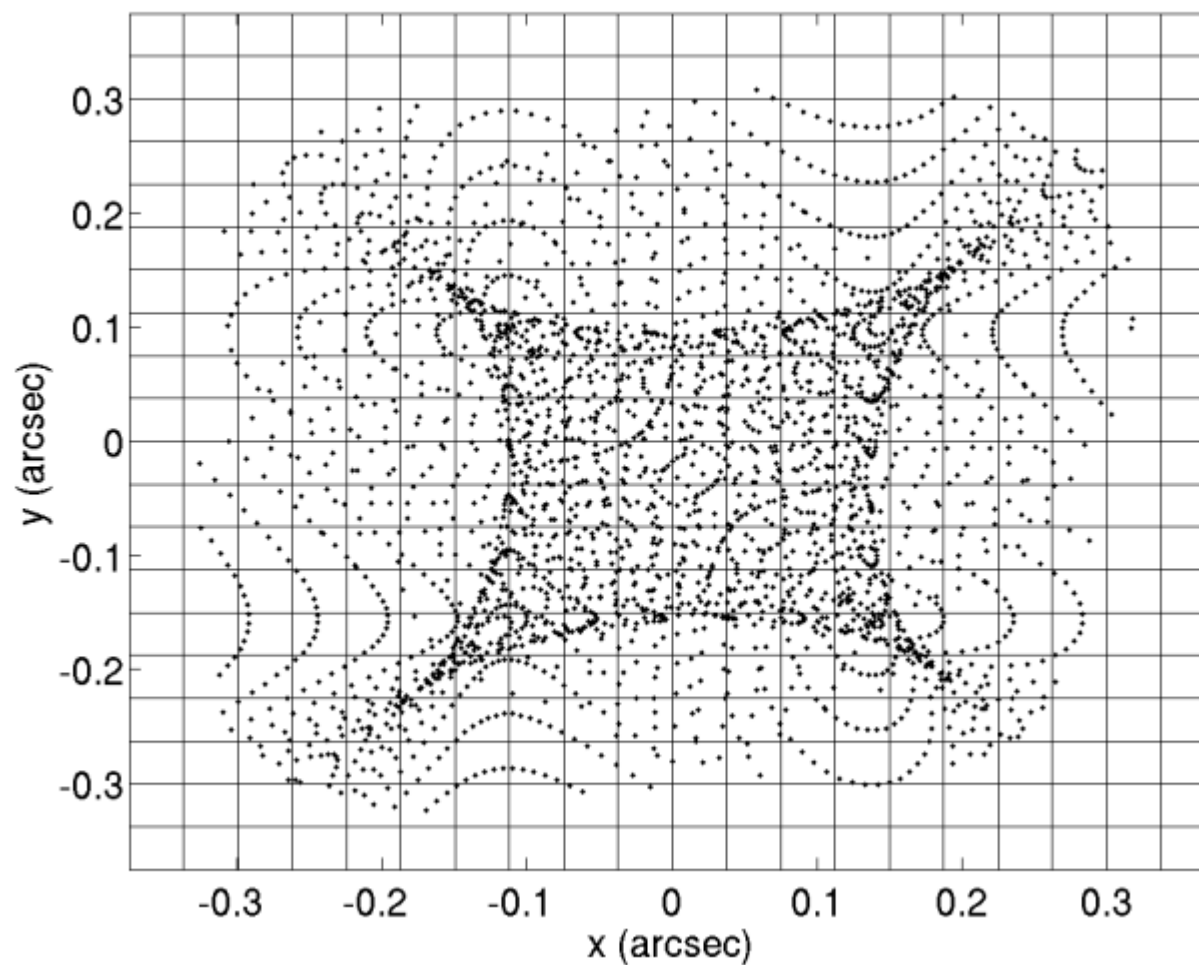
Source Plane - Phase 1





Phase Shifts

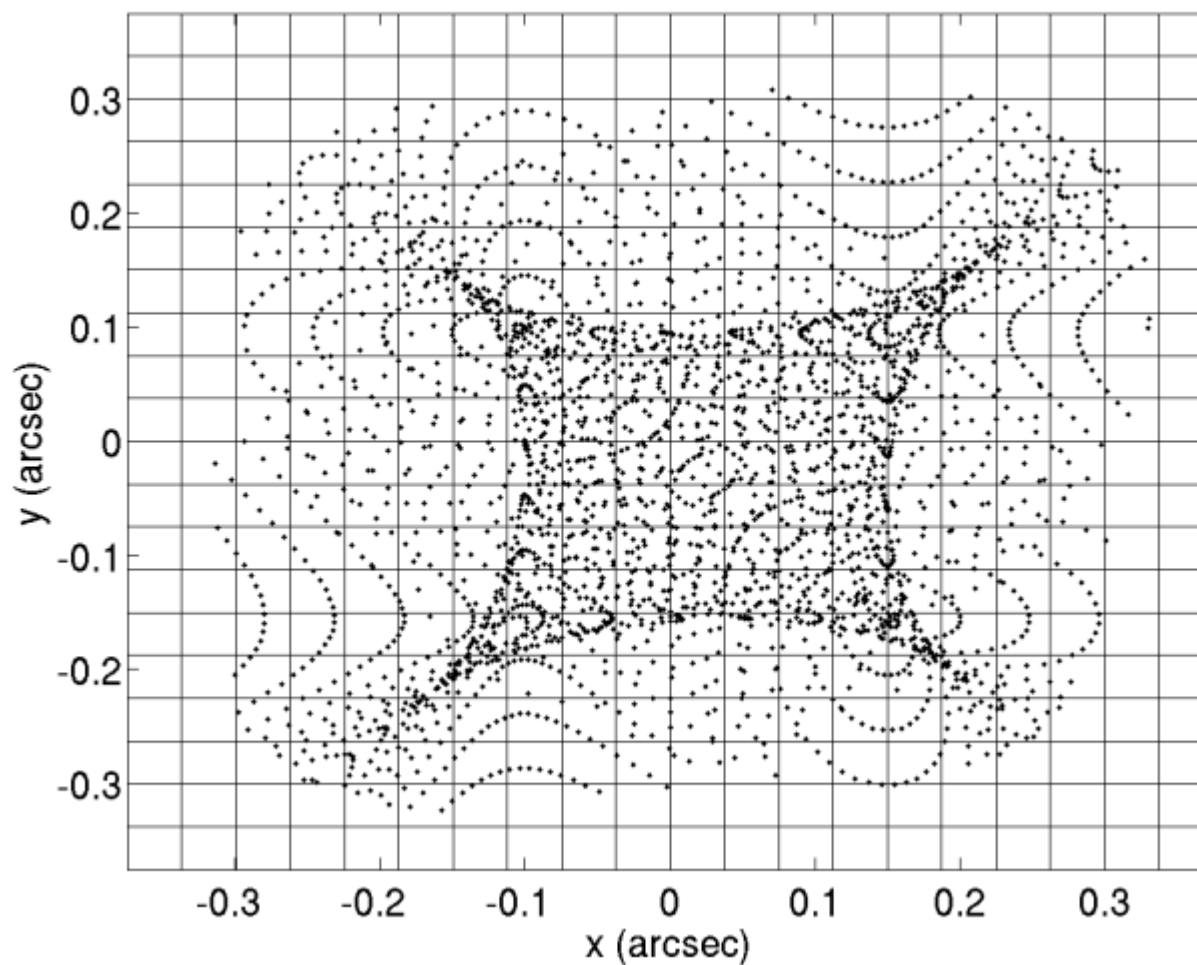
Source Plane - Phase 2





Phase Shifts

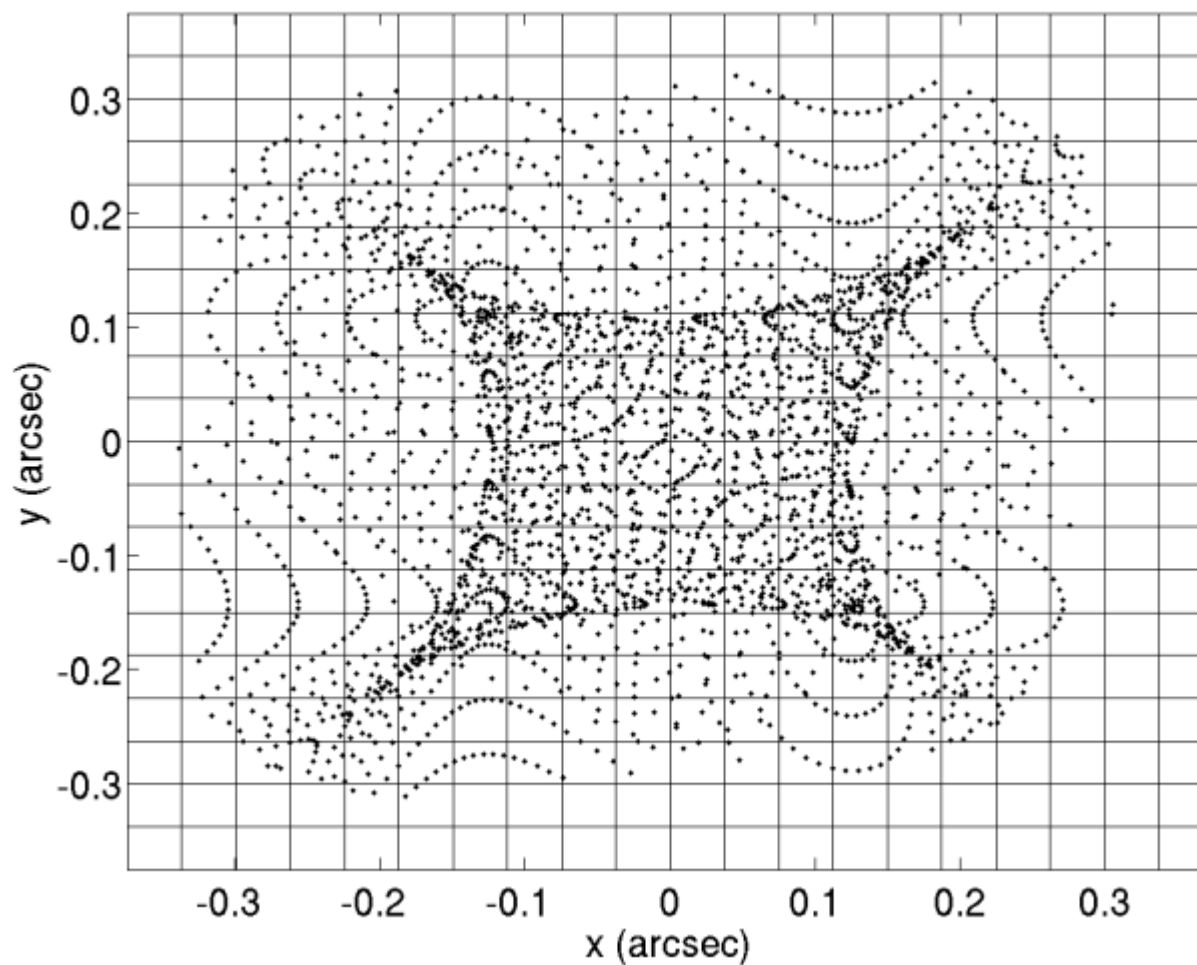
Source Plane - Phase 3





Phase Shifts

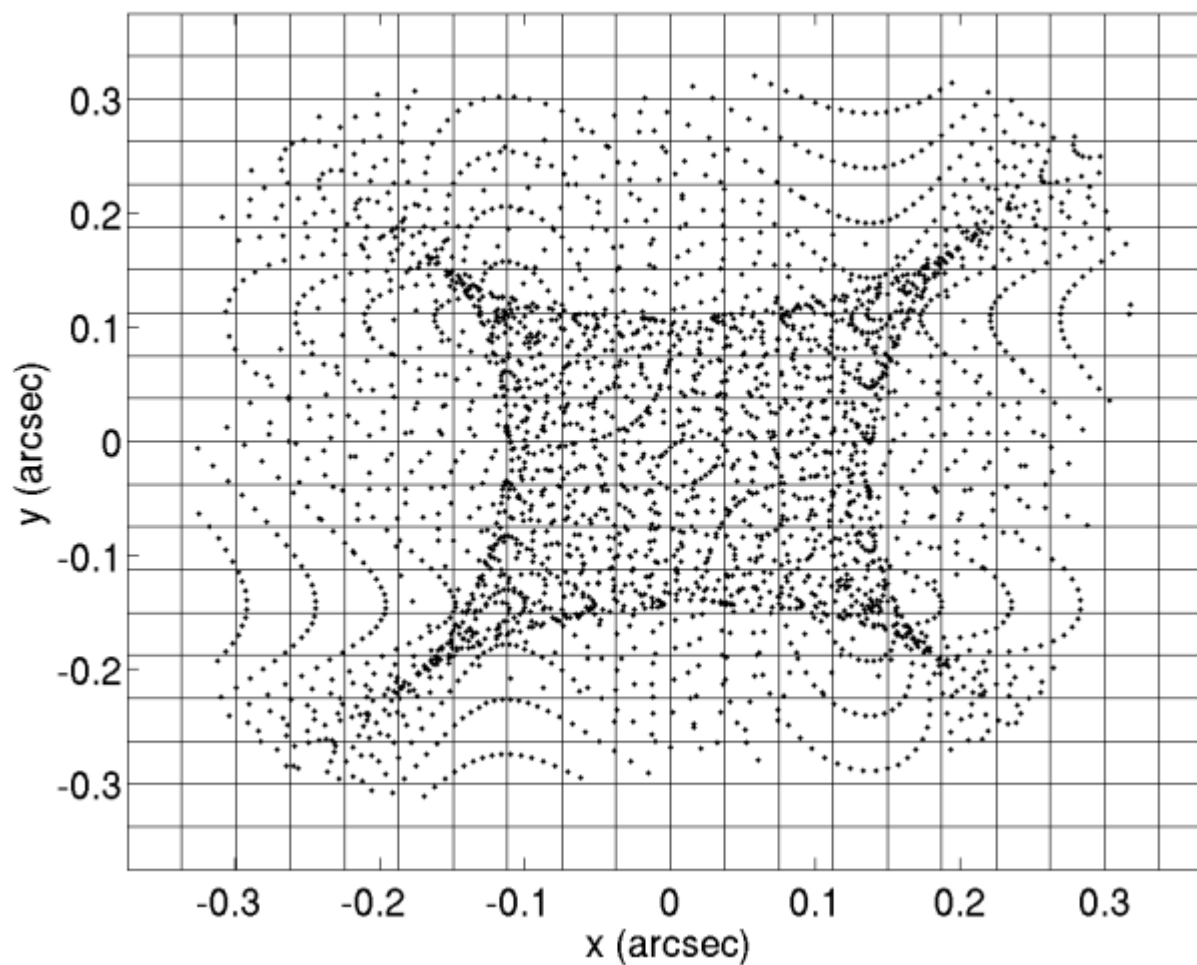
Source Plane - Phase 4





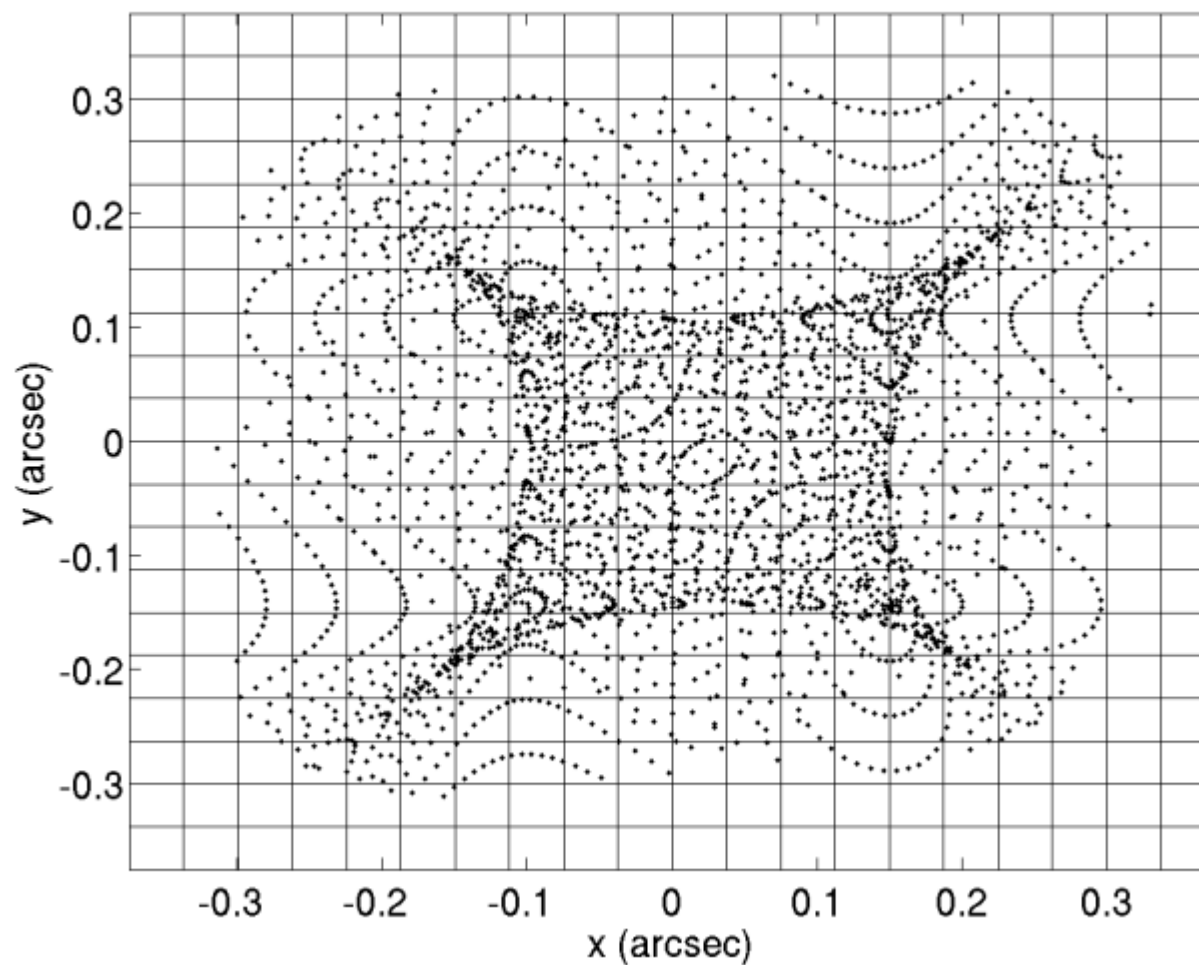
Phase Shifts

Source Plane - Phase 5



Phase Shifts

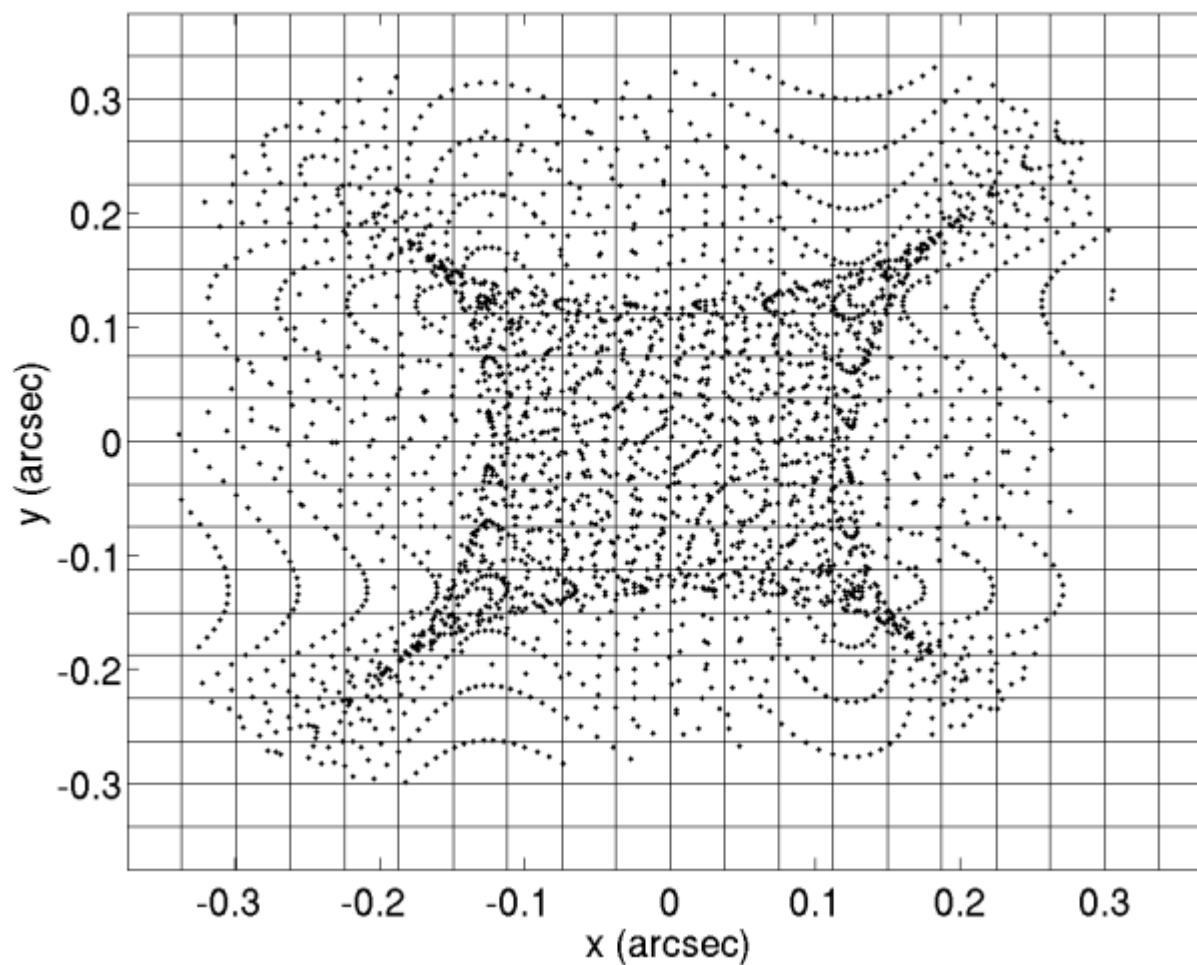
Source Plane - Phase 6





Phase Shifts

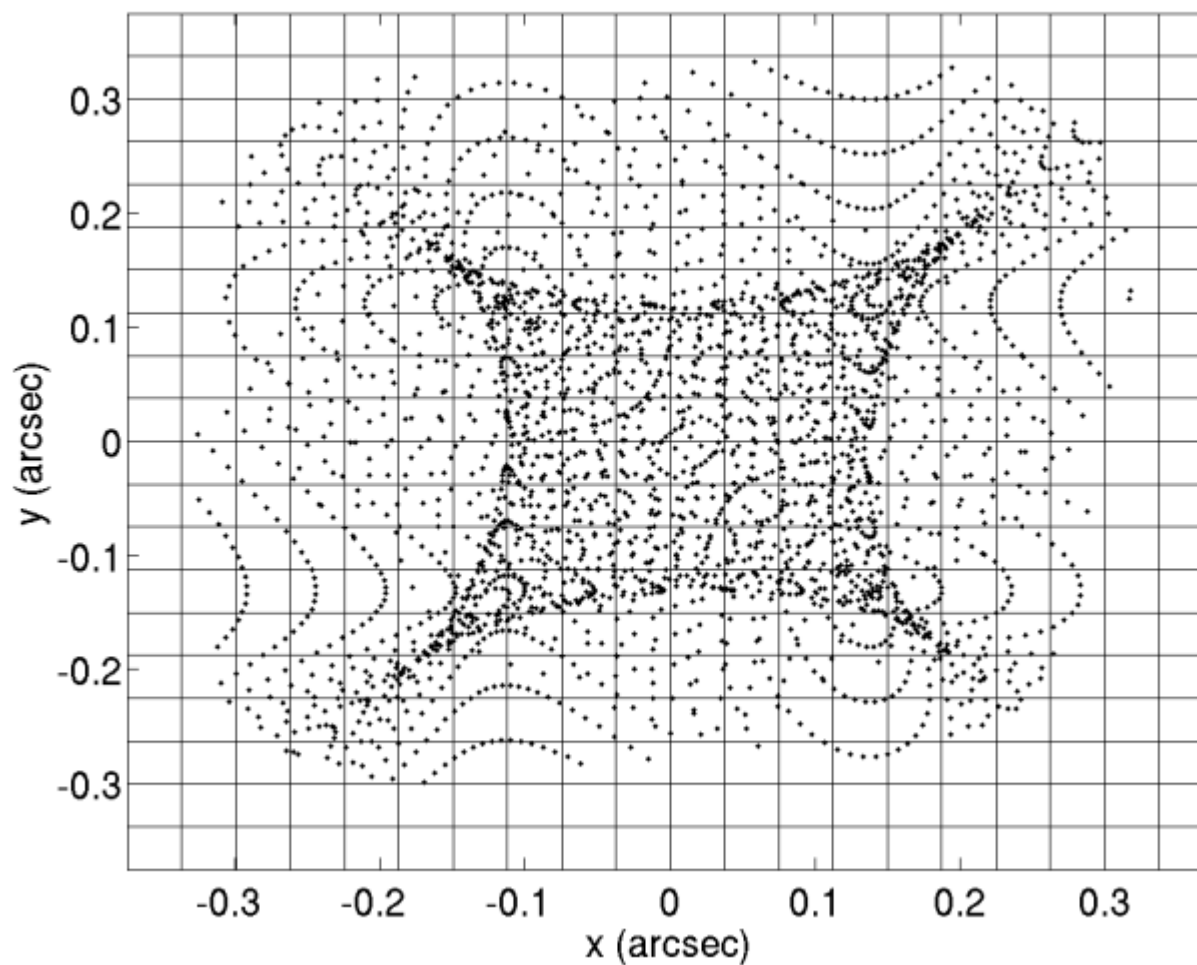
Source Plane - Phase 7





Phase Shifts

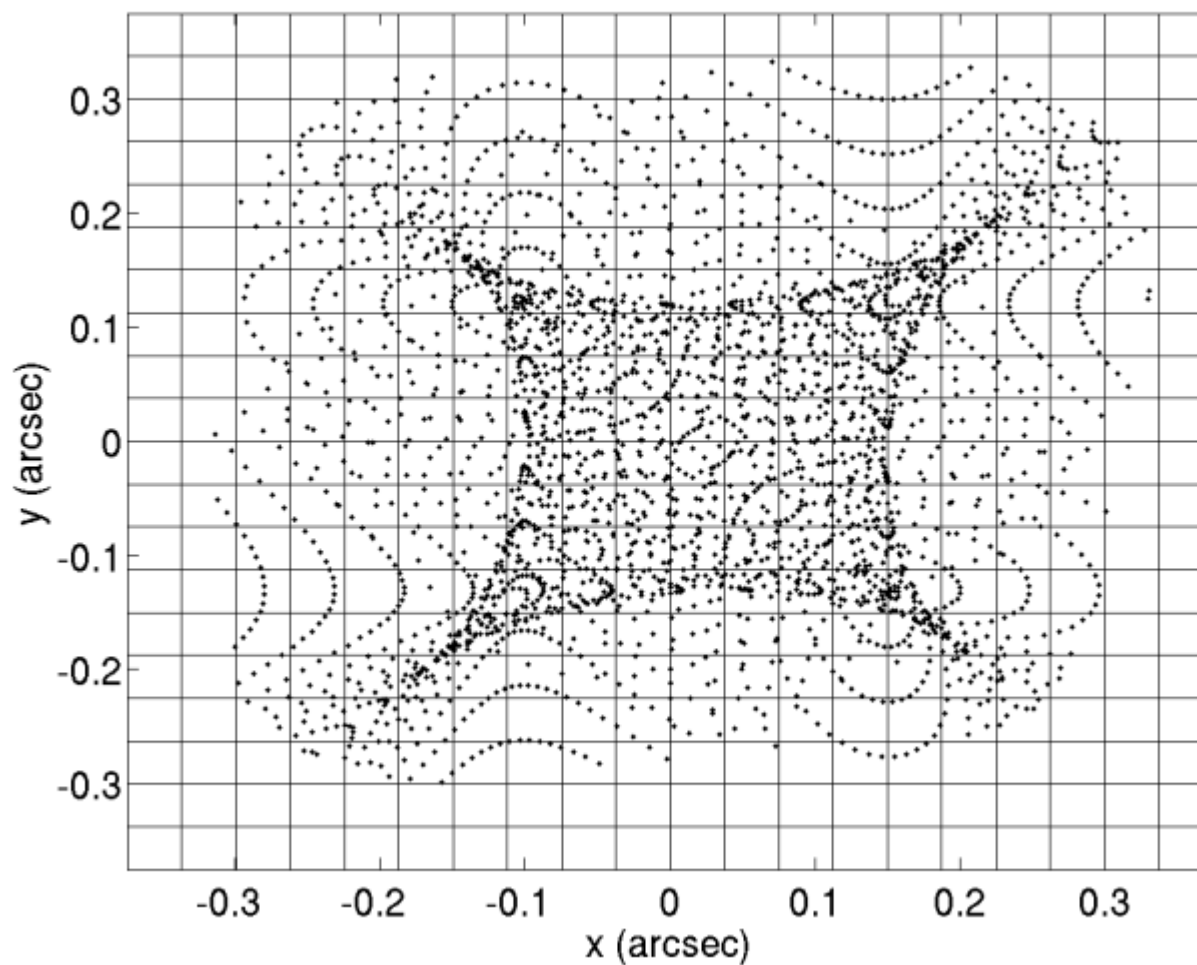
Source Plane - Phase 8





Phase Shifts

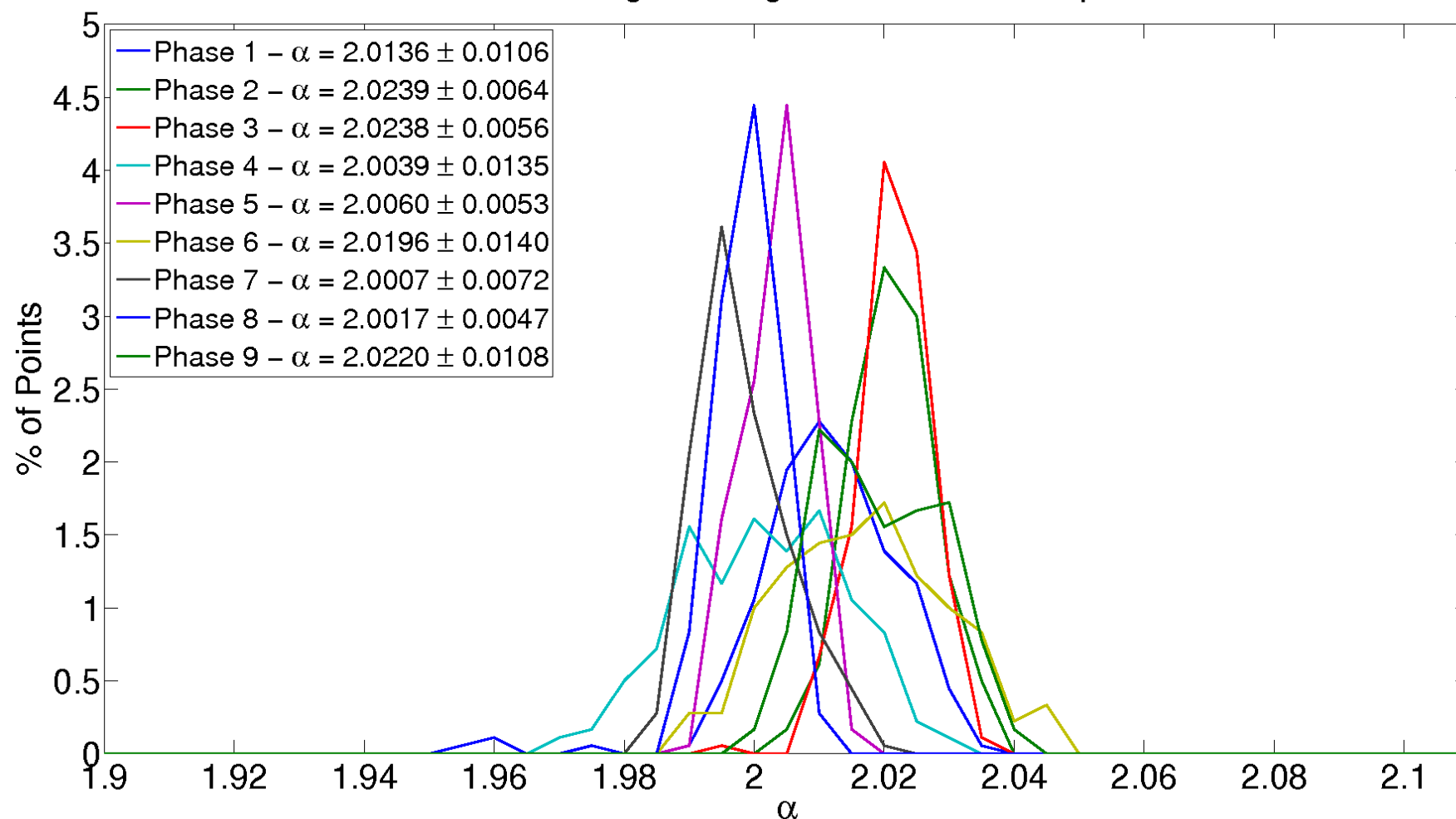
Source Plane - Phase 9





Phase Shifts – α Estimation

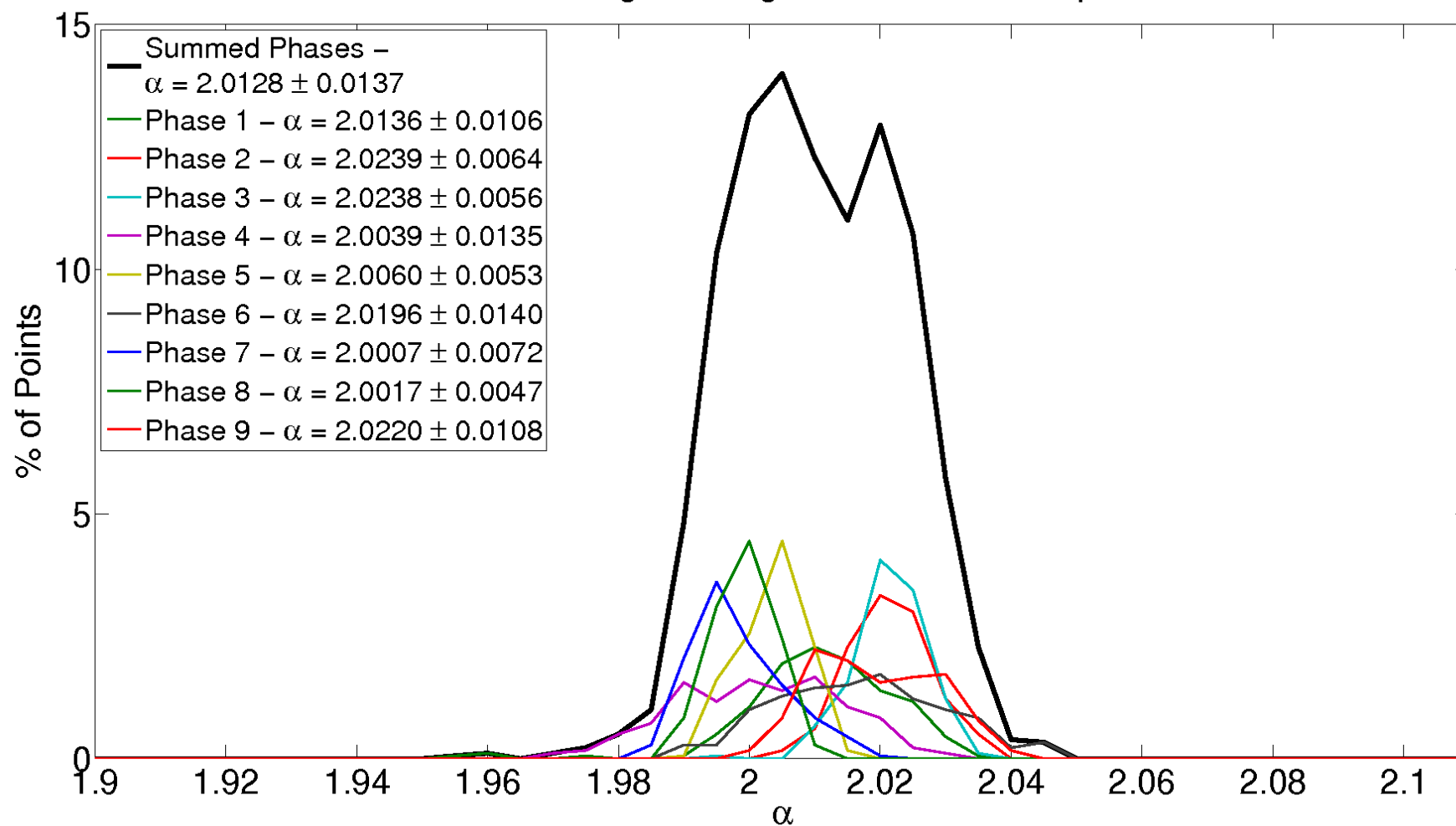
α Estimation of Image 1 using 9 Phase Shifted Square Grids



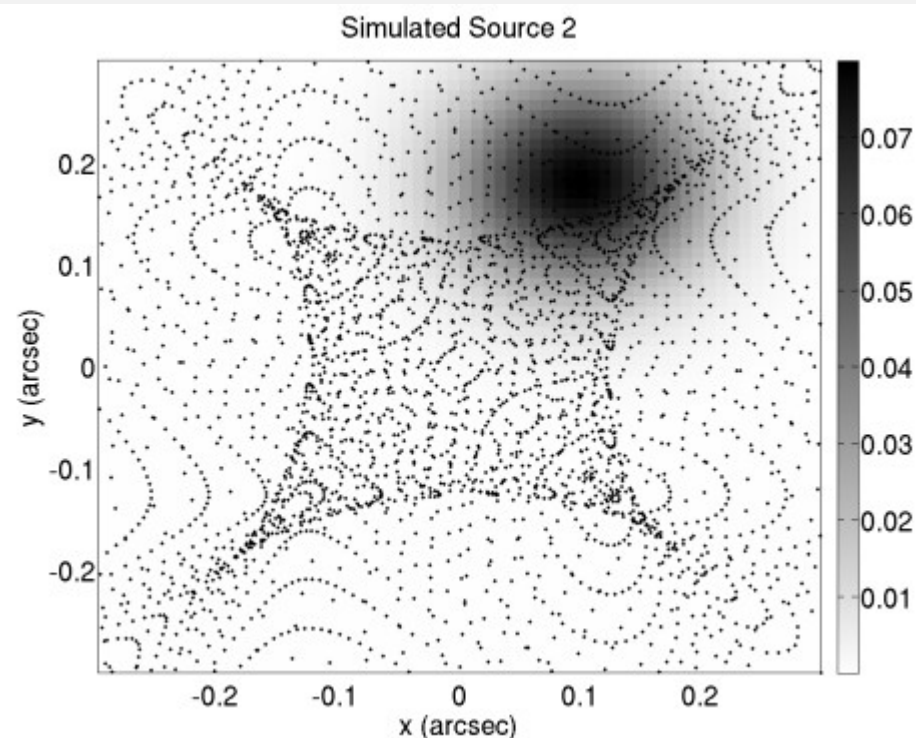
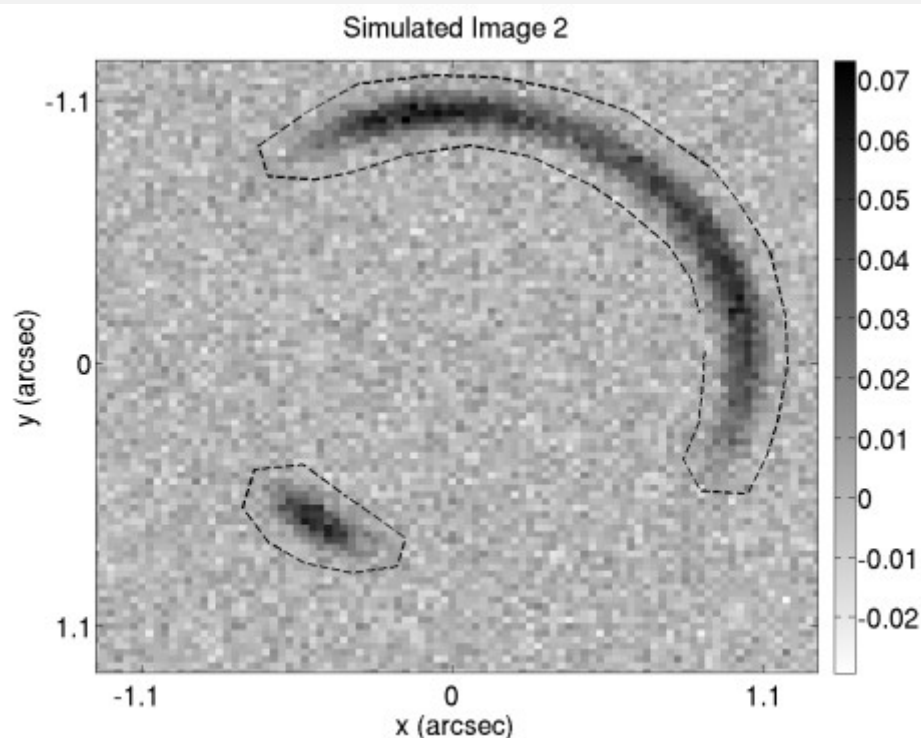


Phase Shifts – α Estimation

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Simulated Image 2



Singular Power-Law Ellipsoid (SPLE) Lens model:

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- $\sigma = 285 \text{ km/s}$

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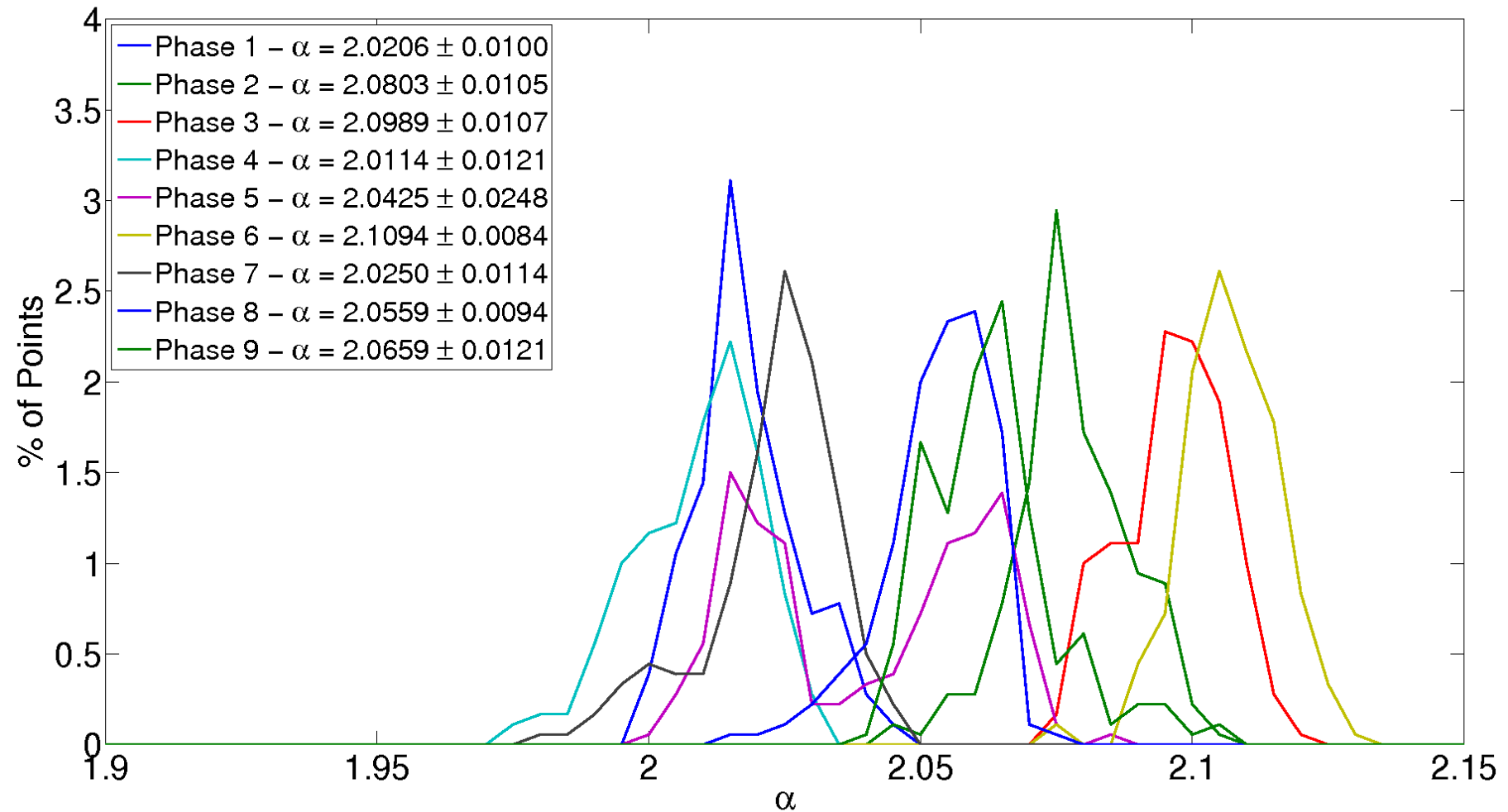
$$\rho(r) \propto r^{-\alpha}$$

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Phase Shifts – α Estimation

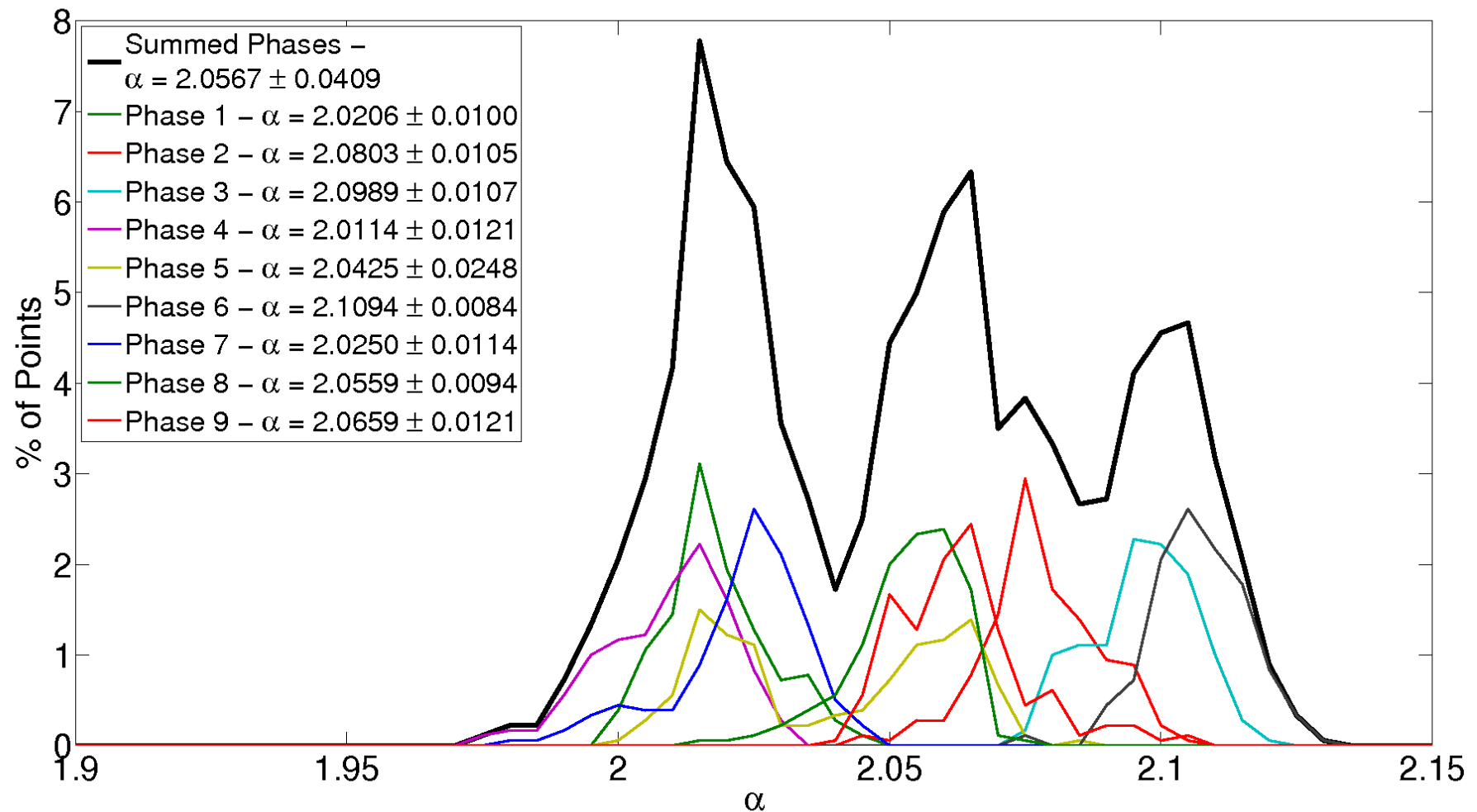
α Estimation of Image 2 using 9 phase shifted Square Grids





Phase Shifts – α Estimation

α Estimation of Image 2 using 9 phase shifted Square Grids





Data Discreteness Bias

- **What is the same in every phase shifted run? - Nearly everything:**
 - Observations – Image, Noise Map, PSF,
 - Inversion Setup – Source Resolution, Image Subgridding, etc
 - Method – Inversion process, Regularization scheme etc.



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- **What is actually changing?**
 - Data Discretization
- **Each phase shifts changes the accessible allocations of image pixels to source pixels.**
 - Directly impacts the evidence calculation for every lens model.



Data Discreteness Bias

- **What does each phase shifts most probable lens model correspond to then?**
 - The lens model where square grid and source reconstruction 'align' favorably.



Data Discreteness Bias

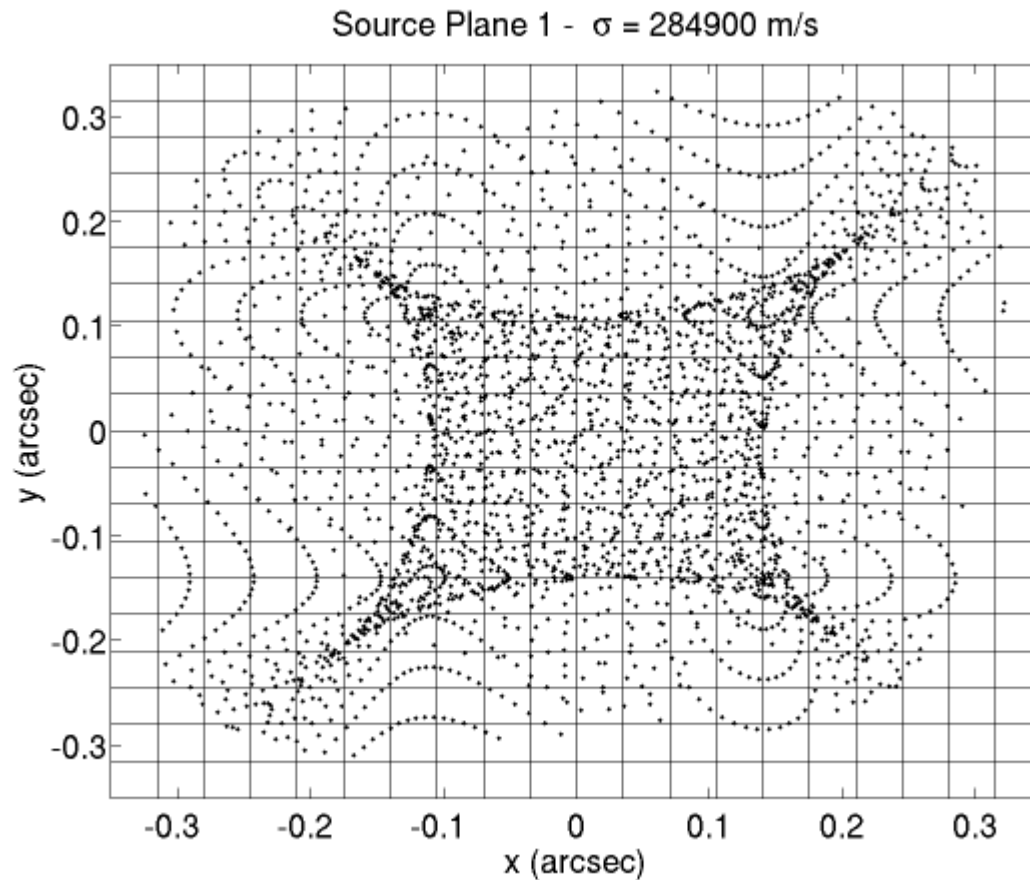
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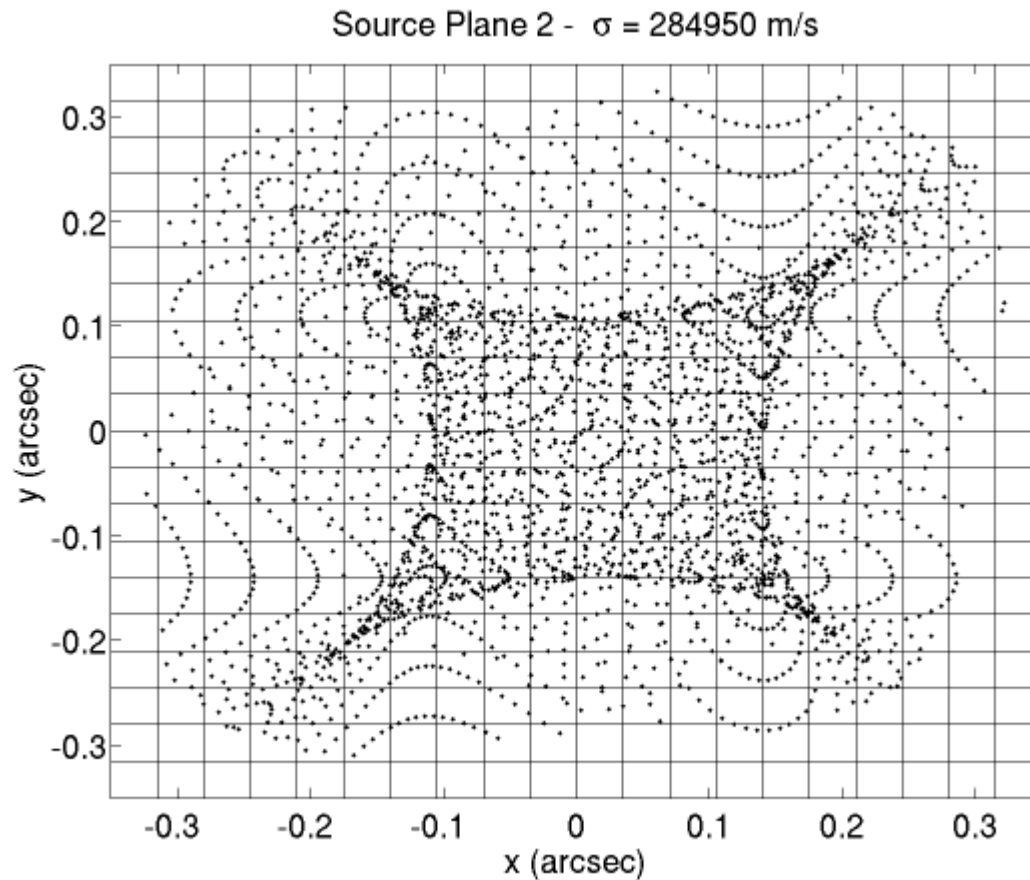
- **What does each phase shifts most probable lens model correspond to?**
 - The lens model where square grid and source reconstruction 'align' favorably.
- **Alignment bias – underlying systematics due to data discretization conspire to give higher evidence.**
- **Not representative of the quality of fit between lens model and observed image!**

Square SLI – Fixed Grid



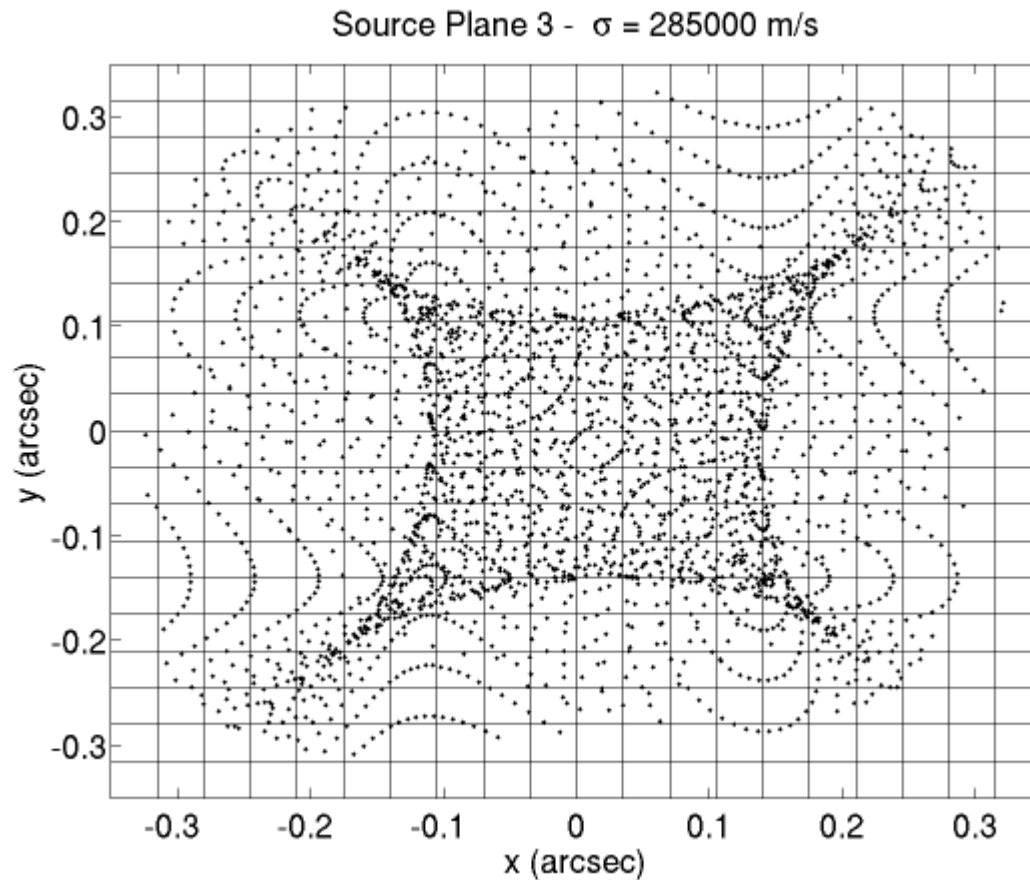
- **5 lens model – all parameters fixed except σ which is changed by 50 m/s in each.**
- $\varepsilon_1 = -1599.877$

Square SLI – Fixed Grid



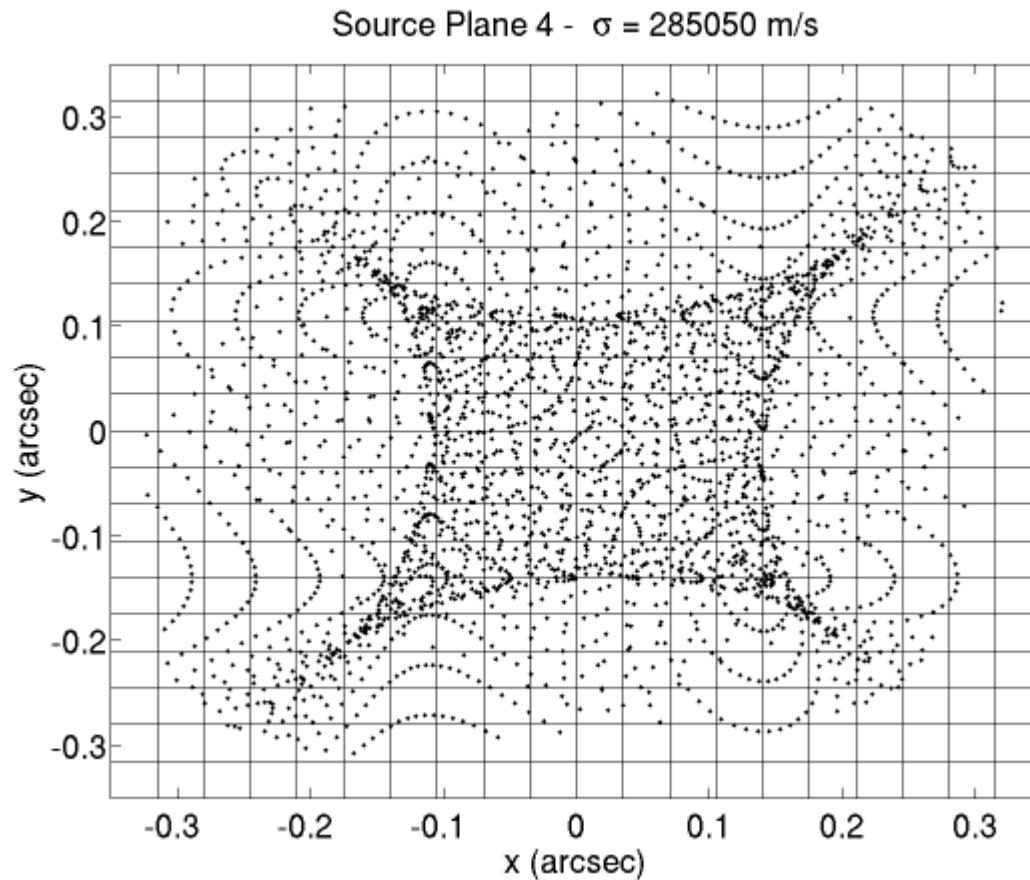
- **5 lens model – all parameters fixed except σ which is changed by 50 m/s in each.**
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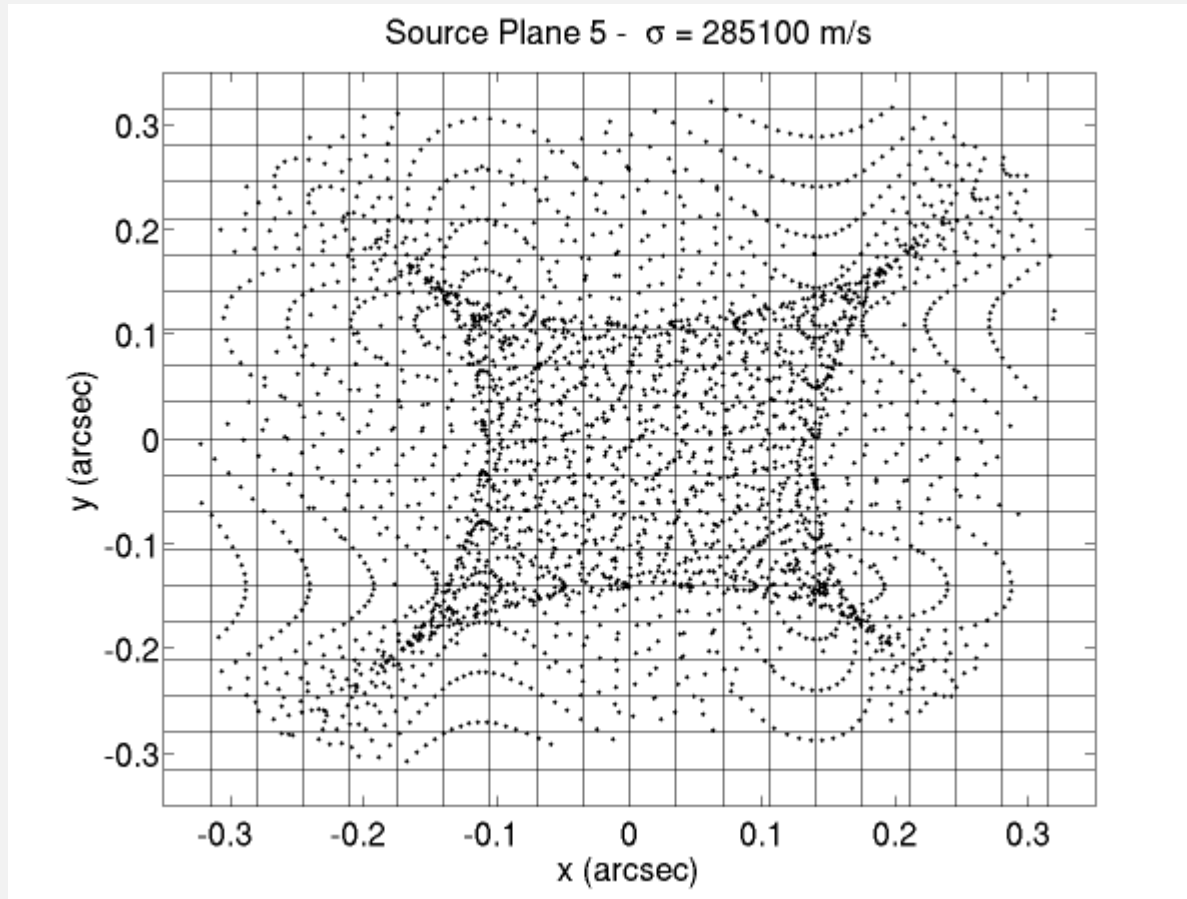
- $\varepsilon_1 = -1599.877$

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- $\varepsilon_4 = -1599.390$

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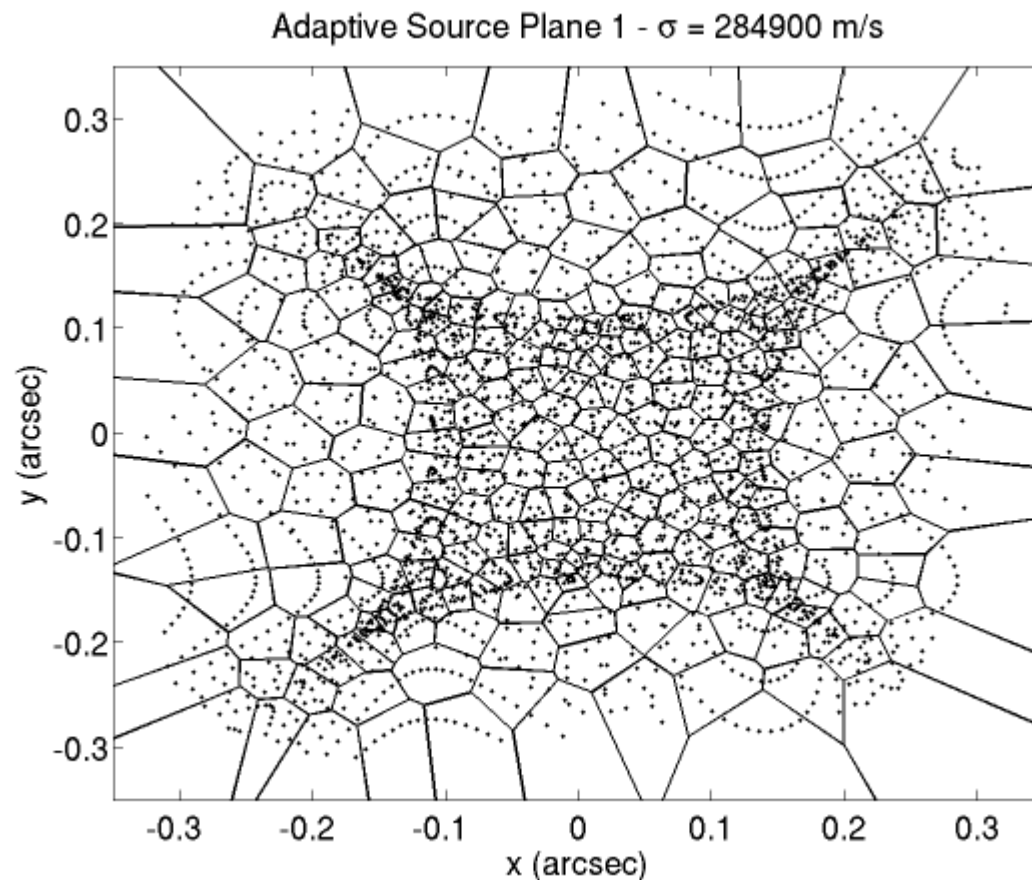
- $\varepsilon_2 = -1599.026$

- $\varepsilon_3 = -1599.000$

- $\varepsilon_4 = -1599.390$

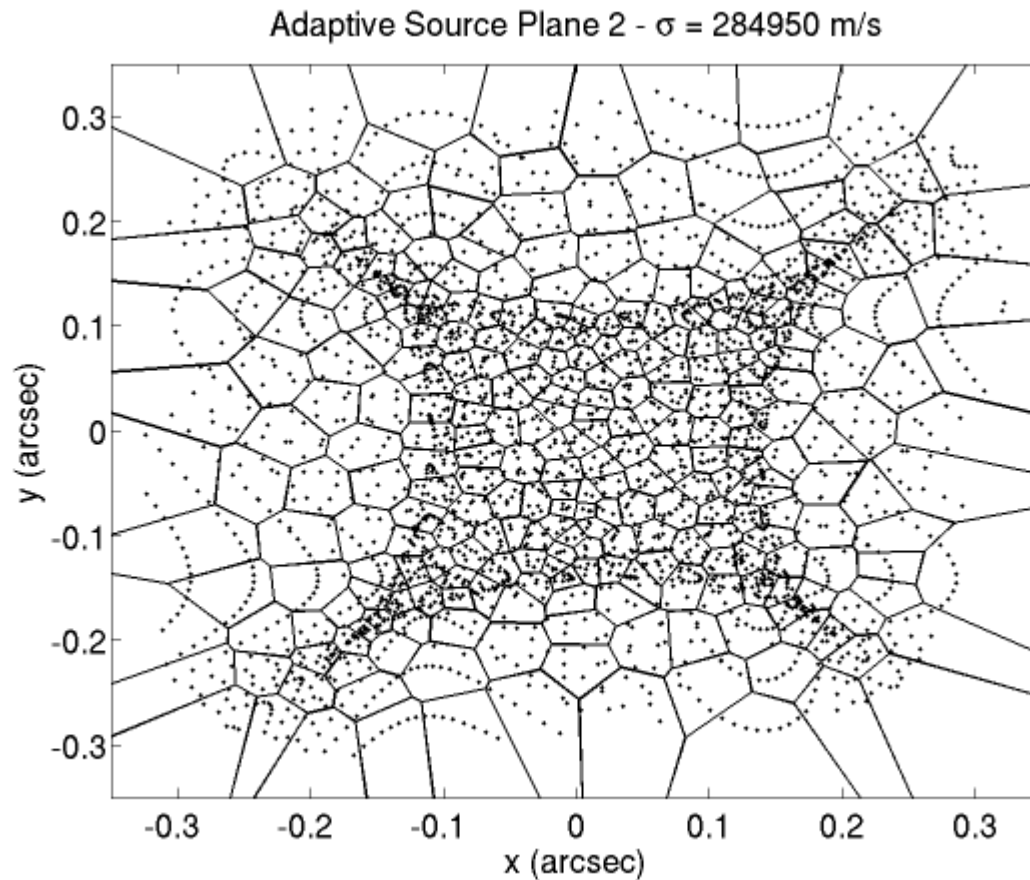
- $\varepsilon_5 = -1599.705$

Adaptive SLI – Unique Grids



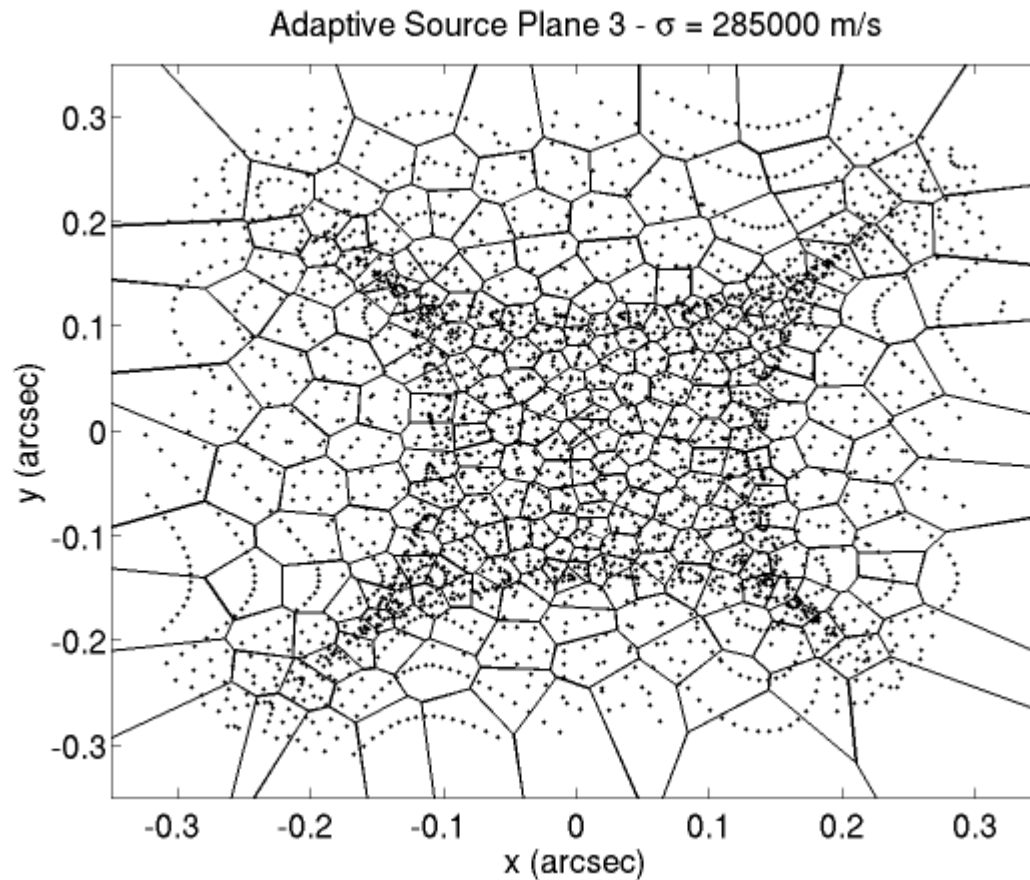
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Adaptive SLI – Unique Grids



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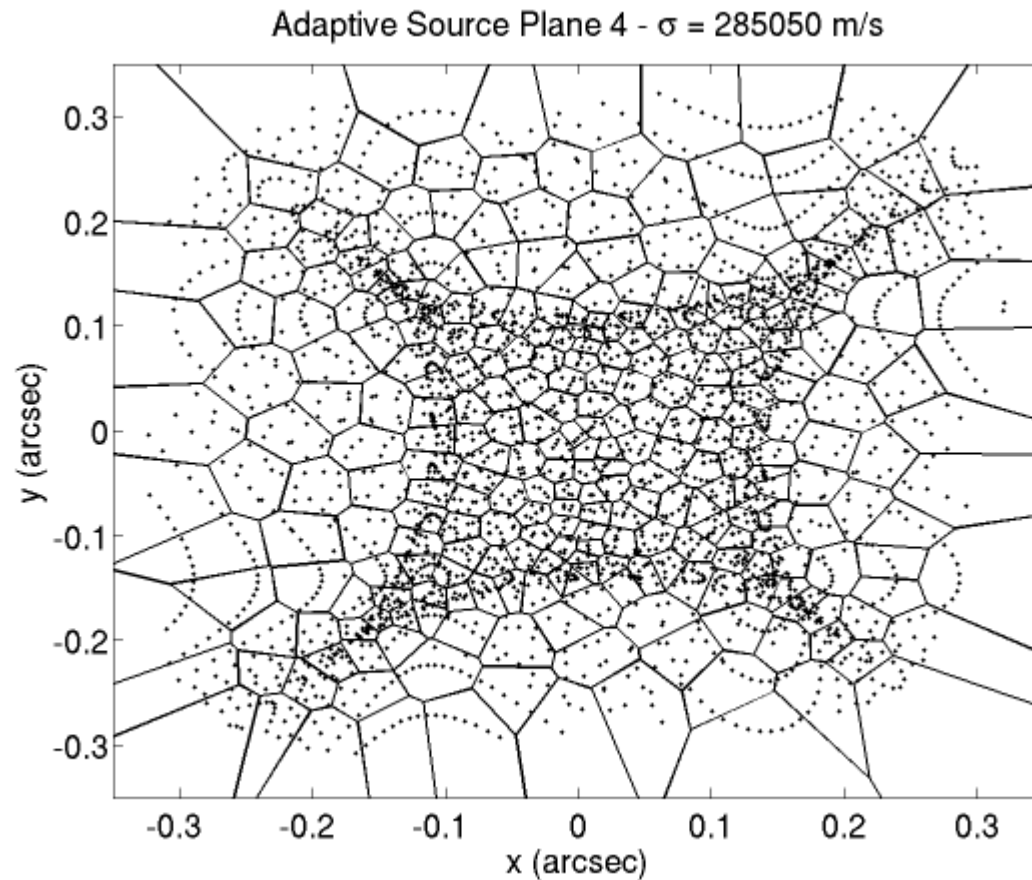
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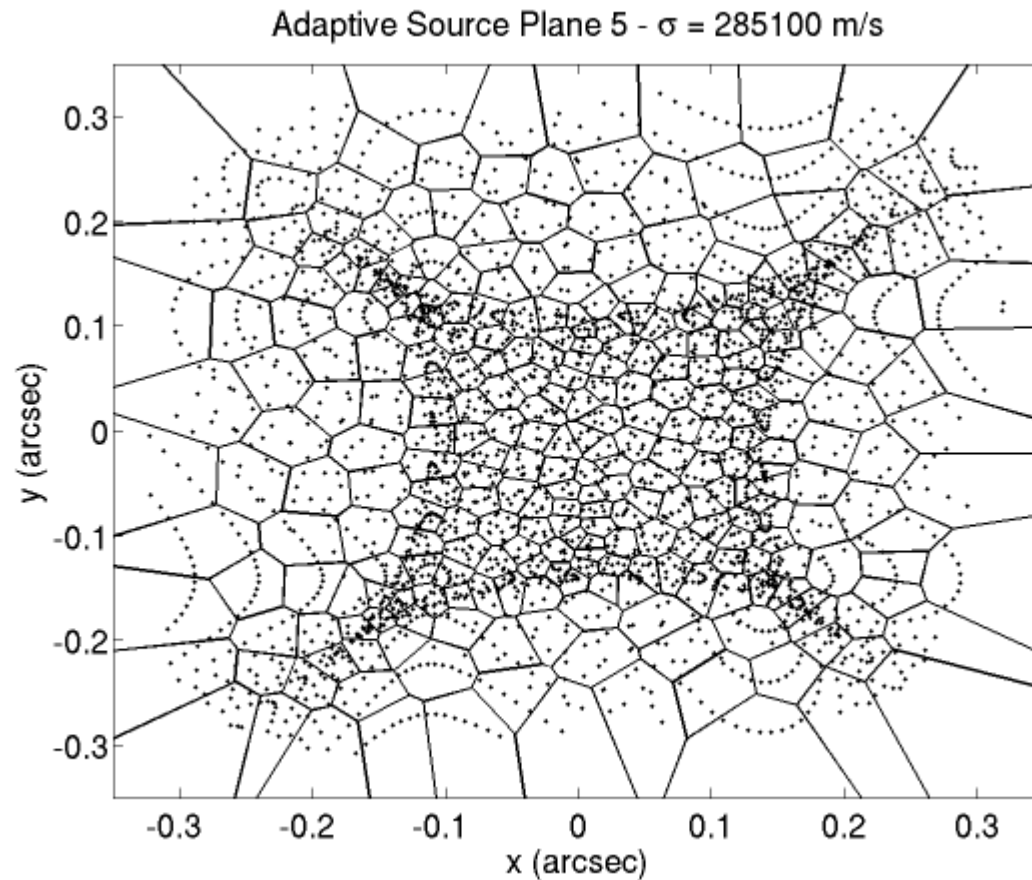
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Adaptive SLI – Unique Grids



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- $\varepsilon_4 = -1604.852$

- $\varepsilon_5 = -1599.654$



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- **Adaptive SLI creates a **unique** grid for every lens model.**
 - Naturally explores effects of changing the data discretization.
 - Removes alignment biases.
- **Adaptive SLI naturally explores all underlying systematics.**
 - Results only dependent on the quality of fit between model and observation.
 - Gives correct errors independent of systematic bias.

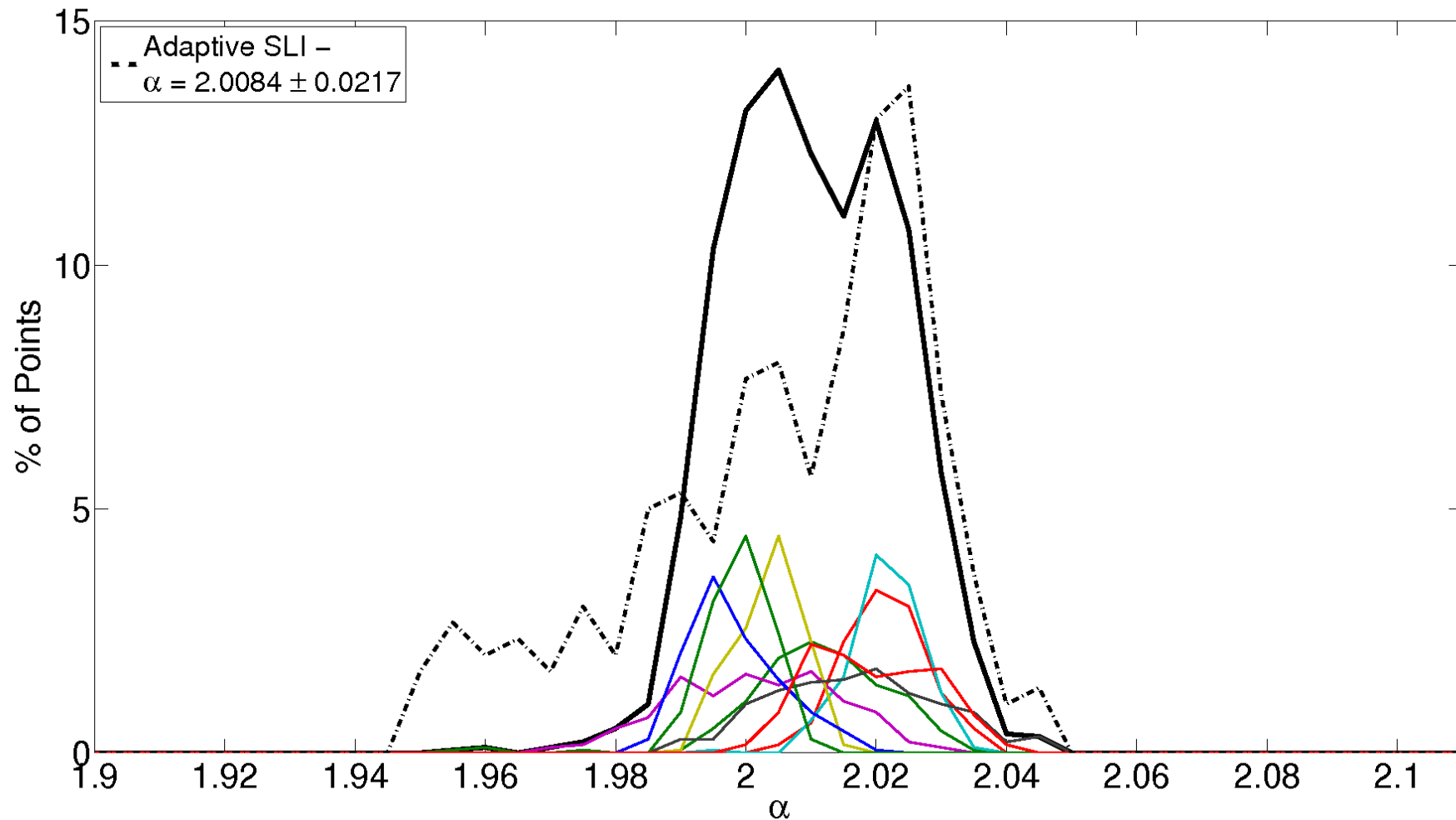


Adaptive SLI - Data Discreteness

- **Adaptive SLI creates a **unique** grid for every lens model.**
 - Naturally explores effects of changing the data discretization.
 - Removes alignment biases.
- **Adaptive SLI naturally explores all underlying systematics.**
 - Results only dependent on the quality of fit between model and observation.
 - Gives correct errors independent of systematic bias.
- **Adaptive SLI creates an **unrelated** grid for every lens model.**
 - Random initialization -> every grid different to one another.
 - Other inversions methods have changing grids, however all have some degree of relatedness.
 - Tests with a fixed initialization -> alignment biases return and errors underestimated.

Adaptive SLI – α Estimation

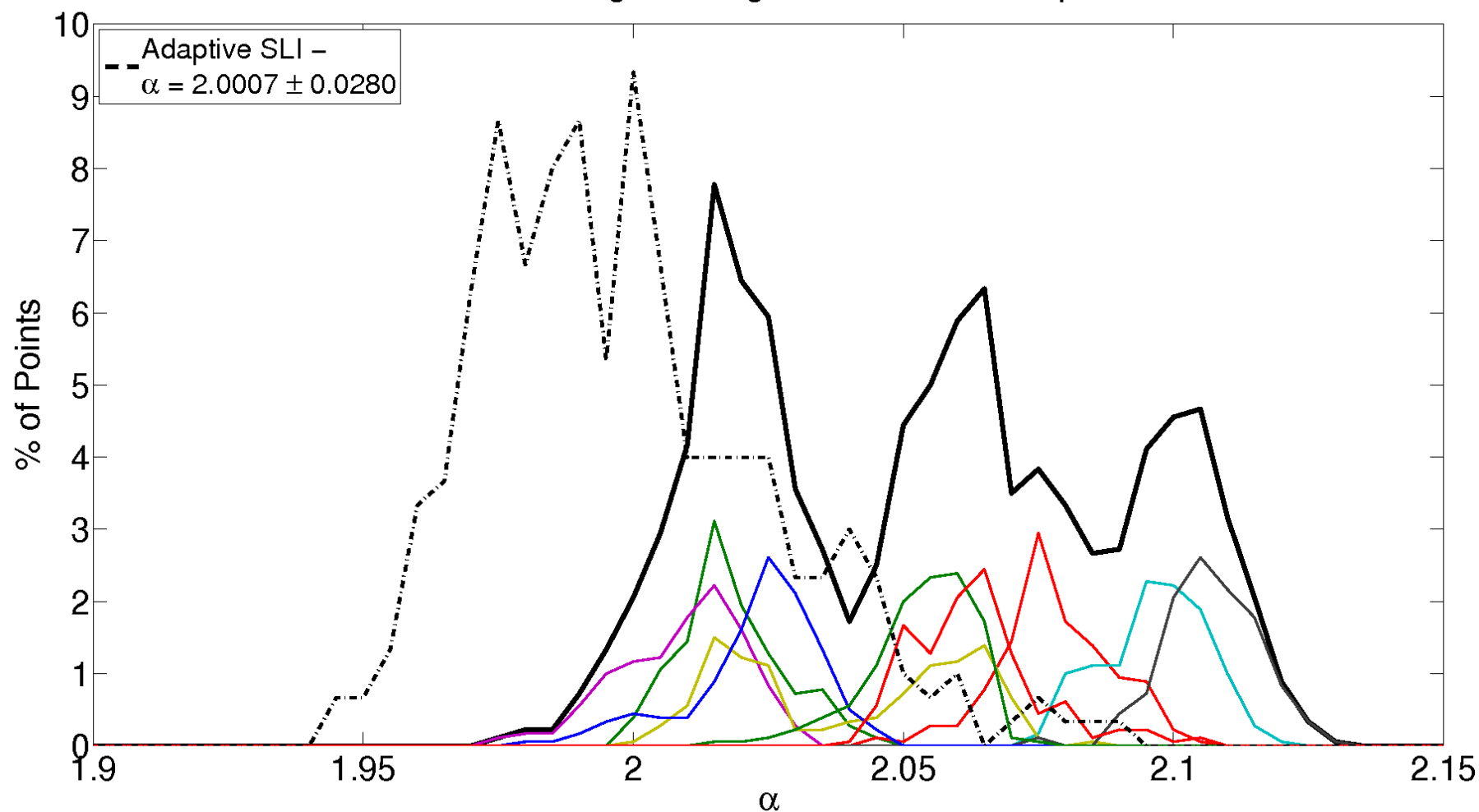
α Estimation of Image 1 using 9 Phase Shifted Square Grids





Adaptive SLI – α Estimation

α Estimation of Image 2 using 9 Phase Shifted Square Grids





Strong Lensing Analysis Pipeline (Preliminary Work)



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 - Minimize user setup pre-inversion -> streamline process.
 - Can use large priors without worry of reducing NDOF.
- **Naturally changes data discretization, removing alignment biases.**
 - Accurately fits degenerate lens models like a PL.
 - Errors determined purely by fit of model and observation.



Pipeline Development – 3 Phase Inversion

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- **Phase 3 – Full Adaptive SLI**
 - Use Adaptive SLI as presented here.
 - Accurate lens model and fully sampled errors.



Pipeline Example – 00472808



Pipeline Development

- **Must rigorously test on simulated data.**